

Model Advisor Report –

sldemo_absbrake.slx

Simulink version: 25.2 Model version: 11.39
System: sldemo_absbrake Current run: 28-Apr-2026 21:55:15
Treat as Referenced Model: off 183 items with a timestamp different than 28-Apr-2026 21:55:15
Configuration File:
Justifications File:

Run Summary

Incomplete		1
Failed		8
Warning		81
Justified		0
Information		0
Passed		192
Not Run		2102
Total		2384

Model Advisor

 By Product 0 0 0 0 0 0 893

 Simulink 0 0 0 0 0 0 44

 *Identify unconnected lines, input ports, and output ports*
Not Run

 *Check root model Inport block specifications*
Not Run

 *Check diagnostic settings ignored during accelerated model reference simulation*
Not Run

 *Check optimization settings*
Not Run

 *Check for parameter tunability information ignored for referenced models*
Not Run

 *Check for implicit signal resolution*
Not Run

 *Check for optimal bus virtuality*
Not Run

 *Check for calls to `slDataTypeAndScale()`*
Not Run

 *Check for Discrete-Time Integrator blocks with initial condition uncertainty*
Not Run

 *Identify disabled library links*
Not Run

 *Identify parameterized library links*
Not Run

 *Identify unresolved library links*
Not Run

 *Identify configurable subsystem blocks in the model for converting to variant subsystem blocks.*
Not Run

 *Check usage of function-call connections*

Not Run

 *Check and update mask image display commands with unnecessary imread() function calls*

Not Run

 *Check and update mask to affirm icon drawing commands dependency on mask workspace*

Not Run

 *Identify Environment Controller blocks to be replaced with Variant Source blocks*

Not Run

 *Identify variant blocks with VariantActivation set to 'inherit from Simulink.VariantControl' but do not use Simulink.VariantControl*

Not Run

 *Check S-functions in the model*

Not Run

 *Runtime diagnostics for S-functions*

Not Run

 *Check if Read/Write diagnostics are enabled for Data Store blocks*

Not Run

 *Check Data Store Memory blocks for multitasking, strong typing, and shadowing issues*

Not Run

 *Check Model History properties*

Not Run

 *Open the Upgrade Advisor*
Not Run

 *Return all simulation outputs as a single Simulink.SimulationOutput object*
Not Run

 *Check structure parameter usage with bus signals*
Not Run

 *Check for large number of function arguments from virtual bus across model reference boundary*
Not Run

 *Check Delay, Unit Delay and Zero-Order Hold blocks for rate transition*
Not Run

 *Check bus signals treated as vectors*
Not Run

 *Check for potentially delayed function-call block return values*
Not Run

 *Identify block output signals with continuous sample time and non-floating point data type*
Not Run

 *Check usage of Merge blocks*
Not Run

 *Check usage of Outport blocks*
Not Run

 *Check usage of Discrete-Time Integrator blocks*
Not Run

 *Check model settings for migration to simplified initialization mode*
Not Run

 *Check for noncontinuous signals that drive derivative ports*
Not Run

 *Check data store block sample times for modeling errors*
Not Run

 *Check for potential ordering issues involving data store access*
Not Run

 *Identify unit mismatches in the model*
Not Run

 *Identify automatic unit conversions in the model*
Not Run

 *Identify disallowed unit systems in the model*
Not Run

 *Identify undefined units in the model*
Not Run

 *Identify ambiguous units in the model*
Not Run

 *Identify questionable operations for strict single-precision design*
Not Run

 Simulink Coder 0 0 0 0 0 0 8

 *Check solver for code generation*

Not Run

 *Check for blocks not supported by code generation*

Not Run

 *Check for model reference configuration mismatch*

Not Run

 *Check code generation identifier formats used for model reference*

Not Run

 *Check for relative execution order change for Data Store Read and Data Store Write blocks*

Not Run

 *Check sample times and tasking mode*

Not Run

 *Check for blocks that have constraints on tunable parameters*

Not Run

 *Check reuse of subsystem code*

Not Run

 Embedded Coder 0 0 0 0 0 0 35

 *Identify lookup table blocks that generate expensive out-of-range checking code*

Not Run

 *Check output types of logic blocks*
Not Run

 *Check the hardware implementation*
Not Run

 *Identify questionable software environment specifications*
Not Run

 *Identify questionable code instrumentation (data I/O)*
Not Run

 *Check configuration parameters for generation of inefficient saturation code*
Not Run

 *Check for blocks not recommended for C/C++ production code deployment*
Not Run

 *Identify blocks generating inefficient algorithms*
Not Run

 *Check configuration parameters for MISRA C:2012*
Not Run

 *Check for blocks not recommended for MISRA C:2012*
Not Run

 *Check for unsupported block names*
Not Run

 *Check usage of Assignment blocks*
Not Run

 *Check for switch case expressions without a default case*
Not Run

 *Check for missing error ports in AUTOSAR receiver interfaces*
Not Run

 *Check for variant blocks that do not have a default choice*
Not Run

 *Check configuration parameters for secure coding standards*
Not Run

 *Check for blocks not recommended for secure coding standards*
Not Run

 *Check signal interfaces*
Not Run

 *Check configuration for component deployment using service interface configuration*
Not Run

 *Check startup and shutdown event*
Not Run

 *Check usage of 'long long' data type*
Not Run

 *Check for blocks not supported for row-major code generation*
Not Run

 *Identify TLC S-Functions with unset array layout*
Not Run

 *Identify blocks that generate expensive fixed-point and saturation code*
Not Run

 *Check for missing const qualifiers in model functions*
Not Run

 *Check bus object names that are used as bus element names*
Not Run

 *Check modeling style for component deployment*
Not Run

 *Check usage of partial data send*
Not Run

 *Identify questionable subsystem settings*
Not Run

 *Identify questionable fixed-point operations*
Not Run

 *Identify blocks that generate expensive rounding code*
Not Run

 *Check for bitwise operations on signed integers*
Not Run

 *Check for recursive function calls*
Not Run

 *Check for equality and inequality operations on floating-point values*
Not Run

 *Check integer word lengths*
Not Run

 AUTOSAR Blockset 0 0 0 0 0 0 2

 *Check compatibility of AUTOSAR Interpolation Routines*
Not Run

 *Check model configuration parameters for AUTOSAR compliance*
Not Run

 HDL Coder 0 0 0 0 0 0 36

 *Checks for blocks and block settings* 0 0 0 0 0 0 11

 *Check for unsupported blocks*
Not Run

 *Check for MATLAB Function block settings*
Not Run

 *Check for obsolete Unit Delay Enabled/Resettable blocks*
Not Run

 *Check for infinite and continuous sample time sources*
Not Run

 Check for unsupported storage class for signal objects
Not Run

 Check for Stateflow chart settings
Not Run

 Check for Trigonometric Function block for LUT-based approximation method
Not Run

 Identify unconnected lines, input ports, and output ports
Not Run

 Check for HDL Reciprocal block usage
Not Run

 Check for large matrix operations
Not Run

 Check for blocks that have nonzero output latency
Not Run

 *Industry standard checks* 0 0 0 0 0 0 11

 Check architecture name
Not Run

 Check clock settings
Not Run

 Check clock, reset, and enable signals
Not Run

 Check file extension
Not Run

 Check generics
Not Run

 Check naming conventions
Not Run

 Check package file names
Not Run

 Check signal and port names
Not Run

 Check entity and architecture
Not Run

 Check module/entity names
Not Run

 Check top-level subsystem/port names
Not Run

 Model configuration checks 0 0 0 0 0 0 6

 Check delay balancing setting
Not Run

 Check for global reset setting for Xilinx and Altera devices
Not Run

 Check inline configurations setting
Not Run

 Check for model parameters suited for HDL code generation
Not Run

 Check for visualization settings
Not Run

 Check algebraic loops
Not Run

 Checks for ports and subsystems 0 0 0 0 0 0 1

 Check for invalid top level subsystem
Not Run

 Native Floating Point checks 0 0 0 0 0 0 7

 Check for double datatypes in the model
Not Run

 Check for Data Type Conversion blocks with incompatible settings
Not Run

 Check for HDL Reciprocal block usage
Not Run

 Check for Relational Operator block usage
Not Run

 Check for single datatypes in the model
Not Run

 Check for unsupported blocks with Native Floating Point
Not Run

 Check blocks with nonzero ulp error
Not Run

 Simscape 0 0 0 0 0 0 2

 Check consistency of block parameter units
Not Run

 Check for dry hydraulic nodes
Not Run

 Simulink PLC Coder 0 0 0 0 0 0 24

 Industry standard checks 0 0 0 0 0 0 7

 Define the names to avoid
Not Run

 Define the use of case (capitals)
Not Run

 Define the maximum variable name length
Not Run

 Comments must describe the intention of the code
Not Run

 Avoid nested comments

Not Run

 Define maximum number of input/output/in-out variables of a POU

Not Run

 Define type prefixes for variables (if used)

Not Run

 Checks for blocks and block settings  0  0  0  0  0  0  8

 Check if model uses event based blocks

Not Run

 Check if model uses probe blocks

Not Run

 Check if model uses environment controller blocks

Not Run

 Check Stateflow chart update

Not Run

 Check issues with integrator blocks

Not Run

 Check if model can generate testbench

Not Run

 Check function packaging configuration

Not Run

 Check trigonometric blocks

Not Run

 Model configuration checks  0  0  0  0  0  0  9

 Check if model uses blocks that are not supported

Not Run

 Check Data Store Memory blocks

Not Run

 Check model for Stateflow messages

Not Run

 Check if signal lines are configured properly

Not Run

 Check if model uses row major algorithms

Not Run

 Check model mask parameters

Not Run

 Check if model uses machine parented data

Not Run

 Check if model uses custom code

Not Run

 Check model tunable parameters

Not Run

 Simulink 3D Animation 0 0 0 0 0 0 4

 *Check for VR Sink blocks*

Not Run

 *Check for VR Source blocks*

Not Run

 *Check for VR To Video blocks*

Not Run

 *Check for VR RigidBodyTree blocks*

Not Run

 Simulink Check 0 0 0 0 0 0 648

 Modeling Standards 0 0 0 0 0 0 636

 DO-178C/DO-331 Checks 0 0 0 0 0 0 95

 Check usage of standardized MATLAB function headers

Not Run

 Check for MATLAB Function interfaces with inherited properties

Not Run

 Check MATLAB Function metrics

Not Run

 Check MATLAB Code Analyzer messages

Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Check state machine type of Stateflow charts
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check usage of While Iterator blocks
Not Run

 Check usage of For and While Iterator subsystems
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check usage of variant blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check model file name
Not Run

 Check usage of lookup table blocks
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for model elements that do not link to requirements
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check usage of identical modeling patterns
Not Run

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Display model version information
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Metrics for generated code complexity
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 Check usage of Abs blocks
Not Run

 Check usage of For Iterator blocks
Not Run

 Check usage of If blocks and If Action Subsystem blocks
Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check usage of Merge blocks
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of Assignment blocks
Not Run

 Check model element names
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check usage of bit-shift operations
Not Run

 Check for invalid root input and output port connections
Not Run

 Check usage of recursions
Not Run

 Check usage of remainder and reciprocal operations
Not Run

 Check usage of square root operations
Not Run

 Check usage of log and log10 operations
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for unreachable and dead code
Not Run

 IEC 61508, IEC 62304, ISO 26262, ISO 25119, EN 50128 and EN 50657 Checks 0 0 0 0 0
0 97

 Check usage of standardized MATLAB function headers
Not Run

 Check for MATLAB Function interfaces with inherited properties
Not Run

 Check MATLAB Function metrics
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Check state machine type of Stateflow charts
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check usage of While Iterator blocks
Not Run

 Check usage of For and While Iterator subsystems
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check usage of variant blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check model file name
Not Run

 Check usage of lookup table blocks
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for model elements that do not link to requirements
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check usage of identical modeling patterns
Not Run

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Display model metrics and complexity report
Not Run

 Check for unconnected objects
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Metrics for generated code complexity
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 Check usage of Abs blocks
Not Run

 Check usage of For Iterator blocks
Not Run

 Check usage of If blocks and If Action Subsystem blocks
Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check usage of Merge blocks
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of Assignment blocks
Not Run

 Check model element names
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check usage of bit-shift operations
Not Run

 Check for invalid root input and output port connections
Not Run

 Display configuration management data
Not Run

 Check usage of recursions
Not Run

 Check usage of remainder and reciprocal operations
Not Run

 Check usage of square root operations
Not Run

 Check usage of log and log10 operations
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for unreachable and dead code
Not Run

 Check for prohibited sink blocks

Not Run

 Check model diagnostic parameters

Not Run

 Check signal line labels

Not Run

 Check for propagated signal labels

Not Run

 Check lines of code in MATLAB Functions

Not Run

 Check whether block names appear below blocks

Not Run

 Check for mixing basic blocks and subsystems

Not Run

 Check usage of tunable parameters in blocks

Not Run

 Check Implement logic signals as Boolean data (vs. double)

Not Run

 Check the display attributes of block names

Not Run

 Check display for port blocks
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check for display of block parameter
Not Run

 Check return value assignments in Stateflow graphical functions
Not Run

 Check for pointers in Stateflow charts
Not Run

 Check logical expressions in If blocks
Not Run

 Check for Simulink diagrams using nonstandard display attributes
Not Run

 Check input and output settings of MATLAB Functions
Not Run

 Check MATLAB code for global variables
Not Run

 Check use of Simulink in Stateflow charts
Not Run

 Check use of default variants
Not Run

 Check use of single variable variant conditionals
Not Run

 Check usage of restricted variable names
Not Run

 Check usage of character vector inside MATLAB Function block
Not Run

 Check usage of recommended patterns for Switch/Case statements
Not Run

 Check the number of function calls in MATLAB Function blocks
Not Run

 Check nested conditions in MATLAB Functions
Not Run

 Check default transition placement in Stateflow charts
Not Run

 Check for avoiding algebraic loops between subsystems
Not Run

 Check for cascaded Unit Delay blocks
Not Run

 Check file names
Not Run

 Check port block names
Not Run

 Check subsystem names
Not Run

 Check character usage in block names
Not Run

 Check definition of signal labels
Not Run

 Check Signed Integer Division Rounding mode
Not Run

 Check usage of State names
Not Run

 Check usage of Stateflow comments
Not Run

 Check execution timing for default transition path
Not Run

 Check usage of Merge block
Not Run

 Check usage of internal transitions in Stateflow states
Not Run

 Check usage of transition conditions in Stateflow transitions
Not Run

 Check block orientation
Not Run

 Check usage of parentheses in Stateflow transitions
Not Run

 Check usable number for first index
Not Run

 Check character usage in signal names and bus names
Not Run

 Check uniqueness of Stateflow State, Data and Event names
Not Run

 Check length of model file name
Not Run

 Check length of folder name at every level of model path
Not Run

 Check length of subsystem names
Not Run

 Check length of Inport and Outport names
Not Run

 Check length of signal and bus names
Not Run

 Check length of block names
Not Run

 Check entry formatting in State blocks in Stateflow charts
Not Run

 Check prohibited combination of state action and flow chart
Not Run

 Check repetition of Action types
Not Run

 Check for unused data in Stateflow Charts
Not Run

 Check uniqueness of State names
Not Run

 Check if blocks are shaded in the model
Not Run

 Check operator order of Product blocks
Not Run

 Check icon shape of Logical Operator blocks
Not Run

 Check if tunable block parameters are defined as named constants
Not Run

 Check default/else case in Switch Case blocks and If blocks
Not Run

 Check usage of internal transition
Not Run

 Check usage of parallel states
Not Run

 Check scope of data in parallel states
Not Run

 Check indentation of code in Stateflow states
Not Run

 Check for unexpected backtracking in state transitions
Not Run

 Check usage of Lookup Tables
Not Run

 Check for parentheses in Fcn block expressions
Not Run

 Check for unconnected objects in Stateflow Charts
Not Run

 Check duplication of Simulink Data names
Not Run

 Check Model Description
Not Run

 Check Stateflow chart action language
Not Run

 Check character usage in Stateflow data names
Not Run

 Check length of Stateflow data name
Not Run

 Check diagnostic settings for incorrect calculation results
Not Run

 Check usage of transitions to external states
Not Run

 Check order of state action types
Not Run

 Check usage of numeric literals in Stateflow
Not Run

 Check position of comments in transition labels
Not Run

 Check trigger signal names
Not Run

 Check usage of unconditional transitions in flow charts
Not Run

 Check for comments in unconditional transitions
Not Run

 Check terminal junctions in Stateflow
Not Run

 Check if state action type 'exit' is used in the model
Not Run

 Check for consistency in model element names
Not Run

 Check usage of graphical functions in Stateflow
Not Run

 Check for sample time setting
Not Run

 Check usage of Sum blocks
Not Run

 Check Indexing Mode
Not Run

 Check position of Inport and Outport blocks
Not Run

 Check definition of Stateflow events
Not Run

 Check for usage of Data Store Memory blocks
Not Run

 Check signal flow in model
Not Run

 Check position of conditional blocks and iterator blocks
Not Run

 Check Model font settings
Not Run

 Check usage of Simulink functions in Stateflow
Not Run

 Check for exclusive states in state machines
Not Run

 Check for unconnected signal lines and blocks
Not Run

 Check transitions in Stateflow flow charts
Not Run

 Check folder names
Not Run

 Check signal line connections
Not Run

 Check position of signal labels
Not Run

 Check definition of Stateflow data
Not Run

 Check for MATLAB expressions in Stateflow blocks
Not Run

 Check Stateflow transition appearance
Not Run

 Check Signal name propagation
Not Run

 Check usage of Discrete-Time Integrator block
Not Run

 Check output data type of operation blocks
Not Run

 Check condition actions and transition actions in Stateflow
Not Run

 Check position of label string in Stateflow transition
Not Run

 Check usage of events in Stateflow charts
Not Run

 Check for missing ports in Variant Subsystems
Not Run

 Check for arrowhead size of transition lines
Not Run

 Check for overlapping or intersecting blocks and signal lines
Not Run

 Check for overlapping of states and transition lines
Not Run

 Check data names in MATLAB Functions
Not Run

 Check the length of data names in MATLAB Functions
Not Run

 Check for size of junctions
Not Run

 Check description of execution statements
Not Run

 Check for spaces in function or variable names
Not Run

 Check for operator precedence
Not Run

 Check spaces in expressions
Not Run

 Check relational operator usage
Not Run

 Check function headers
Not Run

 Check number of lines of functions
Not Run

 Check for utilization of the return value of functions
Not Run

 Check array indices
Not Run

 Check for usage of non-empty statements
Not Run

 Check updates to variables used in state transition conditions
Not Run

 Check for names of Stateflow ports and associated signals
Not Run

 Check usage of Switch blocks
Not Run

 Check usage of unary minus operations in Stateflow charts
Not Run

 Check usage of enumerated values
Not Run

 Check input and output data type for Switch blocks
Not Run

 Check usage of fixed-point data type with non-zero bias
Not Run

 Check signs of input signals in product blocks
Not Run

 Check usage of the Saturation blocks
Not Run

 Check prohibited comparison operation of logical type signals
Not Run

 Check usage of Memory and Unit Delay blocks
Not Run

 Check length of parameter names
Not Run

 Check undefined initial output for conditional subsystems
Not Run

 Check comparison of floating point types in Simulink
Not Run

 Check unused data in Simulink Model
Not Run

 Check for implicit type casting in Stateflow
Not Run

 Check for use of C-style comment symbols
Not Run

 Check Stateflow operators
Not Run

 Check fundamental logical and numerical operations
Not Run

 Check usage of vector and bus signals
Not Run

 Check connections between structural subsystems
Not Run

 Check character usage in parameter names
Not Run

 Check settings for data ports in Multiport Switch blocks
Not Run

 Check type setting by data objects
Not Run

 Check usage of floating-point expressions in Stateflow charts
Not Run

 Check scope of From and Goto blocks
Not Run

 Check Bus and Enumeration data type names
Not Run

 Check length of Bus and Enumeration data type names
Not Run

 Check description of conditional expressions
Not Run

 Check for division by zero in Simulink
Not Run

 JMAAB v5.1 Checks 0 0 0 0 0 0  132

 Check whether block names appear below blocks
Not Run

 Check for mixing basic blocks and subsystems
Not Run

 Check usage of tunable parameters in blocks
Not Run

 Check Implement logic signals as Boolean data (vs. double)
Not Run

 Check the display attributes of block names
Not Run

 Check display for port blocks
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check for display of block parameter
Not Run

 Check return value assignments in Stateflow graphical functions
Not Run

 Check for pointers in Stateflow charts
Not Run

 Check logical expressions in If blocks
Not Run

 Check for Simulink diagrams using nonstandard display attributes
Not Run

 Check input and output settings of MATLAB Functions
Not Run

 Check MATLAB code for global variables
Not Run

 Check use of Simulink in Stateflow charts
Not Run

 Check use of default variants
Not Run

 Check use of single variable variant conditionals
Not Run

 Check usage of character vector inside MATLAB Function block
Not Run

 Check for missing ports in Variant Subsystems
Not Run

 Check folder names
Not Run

 Check Signal name propagation
Not Run

 Check updates to variables used in state transition conditions
Not Run

 Check condition actions and transition actions in Stateflow
Not Run

 Check for usage of text inside states
Not Run

 Check position of label string in Stateflow transition
Not Run

 Check output data type of operation blocks
Not Run

 Check position of signal labels
Not Run

 Check for MATLAB expressions in Stateflow blocks
Not Run

 Check definition of Stateflow data
Not Run

 Check Stateflow transition appearance
Not Run

 Check signal line connections
Not Run

 Check usage of events in Stateflow charts
Not Run

 Check default transition placement in Stateflow charts
Not Run

 Check for avoiding algebraic loops between subsystems
Not Run

 Check for cascaded Unit Delay blocks
Not Run

 Check file names
Not Run

 Check port block names
Not Run

 Check subsystem names
Not Run

 Check character usage in block names
Not Run

 Check definition of signal labels
Not Run

 Check Signed Integer Division Rounding mode
Not Run

 Check usage of State names
Not Run

 Check usage of Stateflow comments
Not Run

 Check execution timing for default transition path
Not Run

 Check usage of Merge block
Not Run

 Check usage of internal transitions in Stateflow states
Not Run

 Check usage of transition conditions in Stateflow transitions
Not Run

 Check block orientation
Not Run

 Check usage of parentheses in Stateflow transitions
Not Run

 Check usable number for first index
Not Run

 Check character usage in signal names and bus names
Not Run

 Check uniqueness of Stateflow State, Data and Event names
Not Run

 Check length of model file name
Not Run

 Check length of folder name at every level of model path
Not Run

 Check length of subsystem names
Not Run

 Check length of Inport and Outport names
Not Run

 Check length of signal and bus names
Not Run

 Check length of block names
Not Run

 Check entry formatting in State blocks in Stateflow charts
Not Run

 Check prohibited combination of state action and flow chart
Not Run

 Check repetition of Action types
Not Run

 Check for unused data in Stateflow Charts
Not Run

 Check uniqueness of State names
Not Run

 Check if blocks are shaded in the model
Not Run

 Check operator order of Product blocks
Not Run

 Check icon shape of Logical Operator blocks
Not Run

 Check if tunable block parameters are defined as named constants
Not Run

 Check default/else case in Switch Case blocks and If blocks
Not Run

 Check usage of internal transition
Not Run

 Check usage of parallel states
Not Run

 Check scope of data in parallel states
Not Run

 Check indentation of code in Stateflow states
Not Run

 Check for unexpected backtracking in state transitions
Not Run

 Check usage of Lookup Tables
Not Run

 Check for parentheses in Fcn block expressions
Not Run

 Check for unconnected objects in Stateflow Charts
Not Run

 Check duplication of Simulink Data names
Not Run

 Check Model Description
Not Run

 Check Stateflow chart action language
Not Run

 Check character usage in Stateflow data names
Not Run

 Check length of Stateflow data name
Not Run

 Check diagnostic settings for incorrect calculation results
Not Run

 Check usage of transitions to external states
Not Run

 Check order of state action types
Not Run

 Check usage of numeric literals in Stateflow
Not Run

 Check position of comments in transition labels
Not Run

 Check trigger signal names
Not Run

 Check usage of unconditional transitions in flow charts
Not Run

 Check for comments in unconditional transitions
Not Run

 Check terminal junctions in Stateflow
Not Run

 Check if state action type 'exit' is used in the model
Not Run

 Check for consistency in model element names
Not Run

 Check usage of graphical functions in Stateflow
Not Run

 Check for sample time setting
Not Run

 Check usage of Sum blocks
Not Run

 Check Indexing Mode
Not Run

 Check position of Inport and Outport blocks
Not Run

 Check definition of Stateflow events
Not Run

 Check for usage of Data Store Memory blocks
Not Run

 Check signal flow in model
Not Run

 Check position of conditional blocks and iterator blocks
Not Run

 Check Model font settings
Not Run

 Check usage of Simulink functions in Stateflow
Not Run

 Check for exclusive states in state machines
Not Run

 Check for unconnected signal lines and blocks
Not Run

 Check transitions in Stateflow flow charts
Not Run

 Check usage of Discrete-Time Integrator block
Not Run

 Check usage of floating-point expressions in Stateflow charts
Not Run

 Check scope of From and Goto blocks
Not Run

 Check usage of Switch blocks
Not Run

 Check usage of unary minus operations in Stateflow charts
Not Run

 Check usage of enumerated values
Not Run

 Check settings for data ports in Multiport Switch blocks
Not Run

 Check type setting by data objects
Not Run

 Check character usage in parameter names
Not Run

 Check input and output data type for Switch blocks
Not Run

 Check usage of fixed-point data type with non-zero bias
Not Run

 Check signs of input signals in product blocks
Not Run

 Check usage of the Saturation blocks
Not Run

 Check prohibited comparison operation of logical type signals
Not Run

 Check usage of Memory and Unit Delay blocks
Not Run

 Check length of parameter names
Not Run

 Check undefined initial output for conditional subsystems
Not Run

 Check comparison of floating point types in Simulink
Not Run

 Check unused data in Simulink Model
Not Run

 Check for implicit type casting in Stateflow
Not Run

 Check for use of C-style comment symbols
Not Run

 Check Stateflow operators
Not Run

 Check fundamental logical and numerical operations
Not Run

 Check usage of vector and bus signals
Not Run

 Check connections between structural subsystems
Not Run

 Check for division by zero in Simulink
Not Run

 JMAAB v6 Checks  0  0  0  0  0  0  152

 Check whether block names appear below blocks
Not Run

 Check for mixing basic blocks and subsystems
Not Run

 Check usage of tunable parameters in blocks
Not Run

 Check Implement logic signals as Boolean data (vs. double)
Not Run

 Check the display attributes of block names
Not Run

 Check display for port blocks
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check for display of block parameter
Not Run

 Check return value assignments in Stateflow graphical functions
Not Run

 Check for pointers in Stateflow charts
Not Run

 Check logical expressions in If blocks
Not Run

 Check for Simulink diagrams using nonstandard display attributes
Not Run

 Check input and output settings of MATLAB Functions
Not Run

 Check MATLAB code for global variables
Not Run

 Check use of Simulink in Stateflow charts
Not Run

 Check use of default variants
Not Run

 Check use of single variable variant conditionals
Not Run

 Check usage of character vector inside MATLAB Function block
Not Run

 Check usage of recommended patterns for Switch/Case statements
Not Run

 Check the number of function calls in MATLAB Function blocks
Not Run

 Check nested conditions in MATLAB Functions
Not Run

 Check default transition placement in Stateflow charts
Not Run

 Check for avoiding algebraic loops between subsystems
Not Run

 Check for cascaded Unit Delay blocks
Not Run

 Check file names
Not Run

 Check port block names
Not Run

 Check subsystem names
Not Run

 Check character usage in block names
Not Run

 Check definition of signal labels
Not Run

 Check Signed Integer Division Rounding mode
Not Run

 Check usage of State names
Not Run

 Check usage of Stateflow comments
Not Run

 Check execution timing for default transition path
Not Run

 Check usage of Merge block
Not Run

 Check usage of internal transitions in Stateflow states
Not Run

 Check usage of transition conditions in Stateflow transitions
Not Run

 Check block orientation
Not Run

 Check usage of parentheses in Stateflow transitions
Not Run

 Check usable number for first index
Not Run

 Check character usage in signal names and bus names
Not Run

 Check uniqueness of Stateflow State, Data and Event names
Not Run

 Check length of model file name
Not Run

 Check length of folder name at every level of model path
Not Run

 Check length of subsystem names
Not Run

 Check length of Inport and Outport names
Not Run

 Check length of signal and bus names
Not Run

 Check length of block names
Not Run

 Check entry formatting in State blocks in Stateflow charts
Not Run

 Check prohibited combination of state action and flow chart
Not Run

 Check repetition of Action types
Not Run

 Check for unused data in Stateflow Charts
Not Run

 Check uniqueness of State names
Not Run

 Check if blocks are shaded in the model
Not Run

 Check operator order of Product blocks
Not Run

 Check icon shape of Logical Operator blocks
Not Run

 Check if tunable block parameters are defined as named constants
Not Run

 Check default/else case in Switch Case blocks and If blocks
Not Run

 Check usage of internal transition
Not Run

 Check usage of parallel states
Not Run

 Check scope of data in parallel states
Not Run

 Check indentation of code in Stateflow states
Not Run

 Check for unexpected backtracking in state transitions
Not Run

 Check usage of Lookup Tables
Not Run

 Check for parentheses inFcn block expressions
Not Run

 Check for unconnected objects in Stateflow Charts
Not Run

 Check duplication of Simulink Data names
Not Run

 Check Model Description
Not Run

 Check Stateflow chart action language
Not Run

 Check character usage in Stateflow data names
Not Run

 Check length of Stateflow data name
Not Run

 Check usage of transitions to external states
Not Run

 Check order of state action types
Not Run

 Check usage of numeric literals in Stateflow
Not Run

 Check position of comments in transition labels
Not Run

 Check trigger signal names
Not Run

 Check usage of unconditional transitions in flow charts
Not Run

 Check for comments in unconditional transitions
Not Run

 Check terminal junctions in Stateflow
Not Run

 Check if state action type 'exit' is used in the model
Not Run

 Check for consistency in model element names
Not Run

 Check usage of graphical functions in Stateflow
Not Run

 Check for sample time setting
Not Run

 Check usage of Sum blocks
Not Run

 Check Indexing Mode
Not Run

 Check position of Inport and Outport blocks
Not Run

 Check definition of Stateflow events
Not Run

 Check for usage of Data Store Memory blocks
Not Run

 Check signal flow in model
Not Run

 Check position of conditional blocks and iterator blocks
Not Run

 Check Model font settings
Not Run

 Check usage of Simulink functions in Stateflow
Not Run

 Check for exclusive states in state machines
Not Run

 Check for unconnected signal lines and blocks
Not Run

 Check transitions in Stateflow flow charts
Not Run

 Check folder names
Not Run

 Check signal line connections
Not Run

 Check position of signal labels
Not Run

 Check definition of Stateflow data
Not Run

 Check for MATLAB expressions in Stateflow blocks
Not Run

 Check Stateflow transition appearance
Not Run

 Check Signal name propagation
Not Run

 Check usage of Discrete-Time Integrator block
Not Run

 Check output data type of operation blocks
Not Run

 Check condition actions and transition actions in Stateflow
Not Run

 Check position of label string in Stateflow transition
Not Run

 Check usage of events in Stateflow charts
Not Run

 Check for missing ports in Variant Subsystems
Not Run

 Check for arrowhead size of transition lines
Not Run

 Check for overlapping or intersecting blocks and signal lines
Not Run

 Check for overlapping of states and transition lines
Not Run

 Check data names in MATLAB Functions
Not Run

 Check the length of data names in MATLAB Functions
Not Run

 Check for size of junctions
Not Run

 Check description of execution statements
Not Run

 Check for spaces in function or variable names
Not Run

 Check for operator precedence
Not Run

 Check spaces in expressions
Not Run

 Check relational operator usage
Not Run

 Check function headers
Not Run

 Check number of lines of functions
Not Run

 Check for utilization of the return value of functions
Not Run

 Check array indices
Not Run

 Check for usage of non-empty statements
Not Run

 Check updates to variables used in state transition conditions
Not Run

 Check usage of Switch blocks
Not Run

 Check usage of unary minus operations in Stateflow charts
Not Run

 Check usage of enumerated values
Not Run

 Check input and output data type for Switch blocks
Not Run

 Check usage of fixed-point data type with non-zero bias
Not Run

 Check signs of input signals in product blocks
Not Run

 Check usage of the Saturation blocks
Not Run

 Check prohibited comparison operation of logical type signals
Not Run

 Check usage of Memory and Unit Delay blocks
Not Run

 Check length of parameter names
Not Run

 Check undefined initial output for conditional subsystems
Not Run

 Check comparison of floating point types in Simulink
Not Run

 Check unused data in Simulink Model
Not Run

 Check for implicit type casting in Stateflow
Not Run

 Check for use of C-style comment symbols
Not Run

 Check Stateflow operators
Not Run

 Check fundamental logical and numerical operations
Not Run

 Check usage of vector and bus signals
Not Run

 Check connections between structural subsystems
Not Run

 Check character usage in parameter names
Not Run

 Check settings for data ports in Multiport Switch blocks
Not Run

 Check type setting by data objects
Not Run

 Check usage of floating-point expressions in Stateflow charts
Not Run

 Check scope of From and Goto blocks
Not Run

 Check Bus and Enumeration data type names
Not Run

 Check length of Bus and Enumeration data type names
Not Run

 Check description of conditional expressions
Not Run

 Check for division by zero in Simulink
Not Run

 *Model Metrics* 0 0 0 0 0 10

 Simulink block metric
Not Run

 Subsystem metric
Not Run

 Library link metric
Not Run

 Effective lines of MATLAB code metric
Not Run

 Stateflow chart objects metric
Not Run

 Lines of code for Stateflow blocks metric
Not Run

 Nondescriptive block name metric
Not Run

 Data and structure layer separation metric
Not Run

 Subsystem depth metric
Not Run

 Cyclomatic complexity metric
Not Run

 Clone Detection and Model Refactoring 0 0 0 0 0 2

 Refactor to simplify bus element blocks
Not Run

 Identify clones from a linked library file
Not Run

 Simulink Code Inspector 0 0 0 0 0 70

 Check code generation settings
Not Run

 Check data import and export settings
Not Run

 *Check diagnostic settings*

Not Run

 *Check hardware implementation settings*

Not Run

 *Check math and data types settings*

Not Run

 *Check solver settings*

Not Run

 *Check for unsupported blocks*

Not Run

 *Check for unconnected objects in the model*

Not Run

 *Check system target file setting*

Not Run

 *Check shared Embedded Code Dictionary configuration*

Not Run

 *Check for unsupported Code Mapping settings*

Not Run

 *Check model arguments for storage classes*

Not Run

 *Check usage of Code in MATLAB Functions*

Not Run

 *Check MATLAB Code Analyzer messages*

Not Run

 *Check storage class for workspace variables*

Not Run

 *Check for unsupported Signal Conversion blocks automatically inserted at signals entering block input ports*

Not Run

 *Check for usage of fixed-point instrumentation*

Not Run

 *Check for usage of synthesized local data stores*

Not Run

 *Check Loop unrolling threshold setting*

Not Run

 *Check usage of global data stores*

Not Run

 *Check global data stores' name shadow*

Not Run

 *Check destinations of If and Switchcase blocks*

Not Run

 *Check for root Outport blocks that have non-auto storage class*

Not Run

 *Check for Terminator blocks that connect to Model block outports*
Not Run

 *Check for unsupported propagation of initial condition values*
Not Run

 *Check data type replacement names*
Not Run

 *Check for multiple sample times in model used as a model reference target*
Not Run

 *Check GetSet storage class for workspace variables*
Not Run

 *Check Treat each discrete rate as a separate task setting*
Not Run

 *Check model for commented out blocks*
Not Run

 *Check model for void_void subsystems that use the same function name*
Not Run

 *Check n-D Lookup Table blocks for incompatible breakpoint data type*
Not Run

 *Check model for reusable subsystems that use the same function interfaces*
Not Run

 *Check for usage of shared synthesized local data stores*
Not Run

 *Check model for compiled and graphical block sorted order*
Not Run

 *Verify 'Generate an example main program' setting of multirate models*
Not Run

 *Check for unsupported usage of 64-bit data types*
Not Run

 *Check for unsupported usage of string data type*
Not Run

 *Check usage of Sources blocks*
Not Run

 *Check usage of Signal Routing blocks*
Not Run

 *Check usage of Math Operations blocks*
Not Run

 *Check usage of Signal Attributes blocks*
Not Run

 *Check usage of Logical and Bit Operations blocks*
Not Run

 *Check usage of Lookup Tables blocks*
Not Run

 *Check usage of User-Defined Function blocks*
Not Run

 *Check usage of Ports and Subsystems blocks*
Not Run

 *Check usage of Discontinuities blocks*
Not Run

 *Check usage of Sinks blocks*
Not Run

 *Check usage of Discrete blocks*
Not Run

 *Check usage of Stateflow blocks*
Not Run

 *Check usage of String blocks*
Not Run

 *Check usage of Stateflow charts*
Not Run

 *Check usage of Stateflow MATLAB action language*
Not Run

 *Check usage of Stateflow transitions*
Not Run

 *Check usage of Stateflow junctions*
Not Run

 *Check usage of Stateflow ports*
Not Run

 *Check usage of Stateflow data*
Not Run

 *Check usage of Stateflow events*
Not Run

 *Check usage of Stateflow states*
Not Run

 *Check usage of Stateflow graphical functions*
Not Run

 *Check usage of Stateflow truth tables*
Not Run

 *Check usage of Stateflow atomic subcharts*
Not Run

 *Check usage of MATLAB Function Blocks*
Not Run

 *Check usage of Data in MATLAB Functions*
Not Run

 *Check usage of root Outport blocks*
Not Run

 *Check usage of buses*
Not Run

 *Check usage of shared utilities*
Not Run

 *Check for sample times in the model*
Not Run

 *Check conditional input branch execution setting*
Not Run

 *Check for unsupported usage of signals in Code Mappings editor*
Not Run

 Requirements Toolbox 0 0 0 0 0 4

 *Requirements Consistency* 0 0 0 0 0 4

 Identify requirement links with missing documents
Not Run

 Identify requirement links that specify invalid locations within documents
Not Run

 Identify selection-based links having description fields that do not match their requirements document text
Not Run

 Identify requirement links with path type inconsistent with preferences
Not Run

 Simulink Design Verifier 0 0 0 0 0 0 15

 *Design Error Detection* 0 0 0 0 0 0 13

 Check for dead logic
Not Run

 Check for out of bound array access
Not Run

 Check for division by zero
Not Run

 Check for integer overflow
Not Run

 Check for non-finite and NaN floating-point values
Not Run

 Check for subnormal floating-point values
Not Run

 Check for specified minimum and maximum value violations
Not Run

 Check for data store access violations
Not Run

 Check for block input range violations
Not Run

 Check usage of remainder and reciprocal operations - hisl_0002
Not Run

 Check usage of square root operations - hisl_0003
Not Run

 Check usage of log and log10 operations - hisl_0004
Not Run

 Check usage of Reciprocal Square Root Blocks - hisl_0028
Not Run

 Check compatibility with Simulink Design Verifier
Not Run

 Check for common defects using Simulink Design Verifier Defect Checker
Not Run

 Simulink Control Design 0 0 0 0 0 1

 Identify time-varying source blocks interfering with frequency response estimation
Not Run

 By Task 1 8 81 0 192 0 1209

 Modeling Physical Systems 0 0 0 1 0 1

 Check consistency of block parameter units (27-Apr-2026 23:28:15)

Identify Simscape blocks with ambiguous setting of parameter units. For example, a block parameter expected in 'Hz' may be specified in the dialog with unit of 'rad/s'. Such settings could lead to unexpected conversion factors applied to the numerical value.

Passed

No Simscape blocks with ambiguous unit setting found in the model.

 *Check for dry hydraulic nodes*

Not Run

 Replacing Blocks That Will Be Removed  0  0  0  0  1  0  0

 *Identify Environment Controller blocks to be replaced with Variant Source blocks (27-Apr-2026 23:28:15)*

Passed

The model does not contain any Environment Controller blocks.

 Inherit from Simulink.VariantControl  0  0  0  0  1  0  0

 *Identify variant blocks with VariantActivation set to 'inherit from Simulink.VariantControl' but do not use Simulink.VariantControl (27-Apr-2026 23:28:15)*

Passed

 Simulink PLC Coder  0  0  16  0  0  0  8

 *Model configuration checks*  0  0  7  0  0  0  1

 *Check Data Store Memory blocks (27-Apr-2026 23:28:15)*

Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check model for Stateflow messages (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if signal lines are configured properly (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model uses row major algorithms (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check model mask parameters (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model uses machine parented data (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model uses custom code (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check model tunable parameters

Not Run

 Checks for blocks and block settings  0  0  8  0  0  1

 Check if model uses event based blocks (27-Apr-2026 23:28:15)

Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model uses probe blocks (27-Apr-2026 23:28:15)

Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model uses environment controller blocks (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check Stateflow chart update (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check issues with integrator blocks (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model uses blocks that are not supported (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check if model can generate testbench (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check function packaging configuration (27-Apr-2026 23:28:15)
Warning

No block selected in the following model

- sldemo_absbrake

Recommended Action

Open Model Advisor with top-level subsystem selected

 Check trigonometric blocks
Not Run

 Industry standard checks  0  0  1  0  0  0  6

 Define the names to avoid (27-Apr-2026 23:28:15)

Names defined as keywords must not be used

Warning

This check is not intended for the root level of the model

Recommended Action

Run this check for the subsystem level

Input Parameters Selection

Name	Value
Open Keywords File	N/A

 Define the use of case (capitals)
Not Run

 Define the maximum variable name length
Not Run

 Comments must describe the intention of the code
Not Run

 Avoid nested comments
Not Run

 Define maximum number of input/output/in-out variables of a POU
Not Run

 Define type prefixes for variables (if used)
Not Run

 S-function Checks 0 0 0 0 0 1

 Check S-functions in the model
Not Run

 Simulation Accuracy 0 0 0 0 0 1

 Check for noncontinuous signals that drive derivative ports
Not Run

 Simulation Runtime Accuracy Diagnostics 0 0 0 0 0 2

 Runtime diagnostics for S-functions
Not Run

 Check if Read/Write diagnostics are enabled for Data Store blocks
Not Run

Managing Data Store Memory Blocks 0 0 0 0 0 0 4

Check Data Store Memory blocks for multitasking, strong typing, and shadowing issues
Not Run

Check data store block sample times for modeling errors
Not Run

Check for potential ordering issues involving data store access
Not Run

Check for relative execution order change for Data Store Read and Data Store Write blocks
Not Run

Simulink Model File Integrity 0 0 0 0 0 0 1

Check Model History properties
Not Run

Model Referencing 0 0 0 0 0 0 8

Check for model reference configuration mismatch
Not Run

Check diagnostic settings ignored during accelerated model reference simulation
Not Run

Check code generation identifier formats used for model reference
Not Run

 *Check for parameter tunability information ignored for referenced models*
Not Run

 *Check for implicit signal resolution*
Not Run

 *Check bus signals treated as vectors*
Not Run

 *Check root model Inport block specifications*
Not Run

 *Check for large number of function arguments from virtual bus across model reference boundary*
Not Run

 Managing Library Links And Variants 0 0 0 0 0 4

 *Identify disabled library links*
Not Run

 *Identify parameterized library links*
Not Run

 *Identify unresolved library links*
Not Run

 *Identify configurable subsystem blocks in the model for converting to variant subsystem blocks.*
Not Run

 Data Transfer Efficiency 0 0 0 0 0 1

 *Check Delay, Unit Delay and Zero-Order Hold blocks for rate transition*
Not Run

 Modeling Signals and Parameters using Buses 0 0 0 0 0 0 3

 *Check for optimal bus virtuality*
Not Run

 *Check structure parameter usage with bus signals*
Not Run

 *Check bus signals treated as vectors*
Not Run

 Code Generation Efficiency 0 0 0 0 0 0 11

 *Check optimization settings*
Not Run

 *Identify questionable software environment specifications*
Not Run

 *Identify lookup table blocks that generate expensive out-of-range checking code*
Not Run

 *Identify questionable code instrumentation (data I/O)*
Not Run

 *Check output types of logic blocks*
Not Run

 *Check configuration parameters for generation of inefficient saturation code*
Not Run

 *Identify blocks that generate expensive rounding code*
Not Run

 *Identify questionable fixed-point operations*
Not Run

 *Check usage of 'long long' data type*
Not Run

 *Identify blocks that generate expensive fixed-point and saturation code*
Not Run

 *Identify blocks generating inefficient algorithms*
Not Run

 Modeling Single-Precision Systems 0 0 0 0 0 1

 *Identify questionable operations for strict single-precision design*
Not Run

 Migrating to Simplified Initialization mode 0 0 0 0 0 4

 *Check usage of Merge blocks*
Not Run

 *Check usage of Outport blocks*
Not Run

 *Check usage of Discrete-Time Integrator blocks*
Not Run

 *Check model settings for migration to simplified initialization mode*
Not Run

 Row-Major Code Generation 0 0 0 0 0 3

 *Identify blocks generating inefficient algorithms*
Not Run

 *Check for blocks not supported for row-major code generation*
Not Run

 *Identify TLC S-Functions with unset array layout*
Not Run

 Modeling Standards for MISRA C:2012 0 0 0 0 0 49

 *Check configuration parameters for MISRA C:2012*
Not Run

 *Check for blocks not recommended for C/C++ production code deployment*
Not Run

 *Check for blocks not recommended for MISRA C:2012*
Not Run

 *Check for unsupported block names*
Not Run

 *Check usage of Assignment blocks*
Not Run

 *Check for switch case expressions without a default case*
Not Run

 *Check for missing error ports in AUTOSAR receiver interfaces*
Not Run

 *Check for bitwise operations on signed integers*
Not Run

 *Check for recursive function calls*
Not Run

 *Check for equality and inequality operations on floating-point values*
Not Run

 *Check for missing const qualifiers in model functions*
Not Run

 *Check integer word lengths*
Not Run

 *Check bus object names that are used as bus element names*
Not Run

 *Check for variant blocks that do not have a default choice*
Not Run

 *High-Integrity Systems* 0 0 0 0 0 0 35

 Check usage of Abs blocks

Not Run

 Check usage of remainder and reciprocal operations

Not Run

 Check usage of square root operations

Not Run

 Check usage of log and log10 operations

Not Run

 Check usage of While Iterator blocks

Not Run

 Check usage of For and While Iterator subsystems

Not Run

 Check usage of For Iterator blocks

Not Run

 Check usage of If blocks and If Action Subsystem blocks

Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks

Not Run

 Check usage of Relational Operator blocks

Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for divide-by-zero calculations
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for unreachable and dead code
Not Run

 Stateflow 0 0 0 0 0 3

 Check Stateflow charts for strong data typing
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 MATLAB 0 0 0 0 0 1

 Check switch statements in MATLAB Function blocks
Not Run

 Configuration 0 0 0 0 0 7

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 0 1

 Check model element names
Not Run

 Modeling Standards for Secure Coding (CERT C, CWE, ISO/IEC TS 17961) 0 0 0 0
0 0 27

 *Check configuration parameters for secure coding standards*
Not Run

 *Check for blocks not recommended for C/C++ production code deployment*
Not Run

 *Check for blocks not recommended for secure coding standards*
Not Run

 *Check usage of Assignment blocks*
Not Run

 *Check for switch case expressions without a default case*
Not Run

 *Check for bitwise operations on signed integers*
Not Run

 *Check for equality and inequality operations on floating-point values*
Not Run

 *Check integer word lengths*
Not Run

 *Check for dead logic*
Not Run

 *Check for integer overflow*
Not Run

 *Check for division by zero*
Not Run

 *Check for out of bound array access*
Not Run

 *Check for specified minimum and maximum value violations*
Not Run

 *High-Integrity Systems*  0  0  0  0  0  0  14

 Simulink 0 0 0 0 9

 Check usage of Abs blocks

Not Run

 Check usage of remainder and reciprocal operations

Not Run

 Check usage of log and log10 operations

Not Run

 Check usage of While Iterator blocks

Not Run

 Check for blocks not recommended for C/C++ production code deployment

Not Run

 Check data types for blocks with index signals

Not Run

 Check usage of Reciprocal Sqrt blocks

Not Run

 Check global variables in graphical functions

Not Run

 Check usage of bit-shift operations

Not Run

 Configuration 0 0 0 0 2

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Naming 0 0 0 0 0 0 2

 Check model file name
Not Run

  Check model element names
Not Run

 Code 0 0 0 0 0 0 1

 Check configuration parameters for MISRA C:2012
Not Run

 Modeling Standards for AUTOSAR 0 0 0 0 0 0 95

 *Check compatibility of AUTOSAR Interpolation Routines*
Not Run

 *Check model configuration parameters for AUTOSAR compliance*
Not Run

 High-Integrity Systems 0 0 0 0 0 0 93

 Check usage of standardized MATLAB function headers
Not Run

 Check for MATLAB Function interfaces with inherited properties
Not Run

 Check MATLAB Function metrics
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Metrics for generated code complexity
Not Run

 Check state machine type of Stateflow charts
Not Run

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 Check usage of Abs blocks
Not Run

 Check usage of square root operations
Not Run

 Check usage of While Iterator blocks
Not Run

 Check usage of For and While Iterator subsystems
Not Run

 Check usage of For Iterator blocks
Not Run

 Check usage of If blocks and If Action Subsystem blocks
Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check safety-related diagnostic settings for data store memory
Not Run

 Check usage of Merge blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of variant blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check model file name
Not Run

 Check model element names
Not Run

 Check usage of lookup table blocks
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check global variables in graphical functions
Not Run

 Check for length of user-defined object names
Not Run

 Check usage of Gain blocks
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for model elements that do not link to requirements
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for invalid root input and output port connections
Not Run

 Check for unreachable and dead code
Not Run

 Check data type of loop control variables
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Check usage of remainder and reciprocal operations
Not Run

 Check usage of log and log10 operations
Not Run

 Upgrading to the Current Simulink Version 0 0 0 0 0 0 1

 *Open the Upgrade Advisor*
Not Run

 Component Deployment Guidelines for Service Interface 0 0 0 0 0 0 5

 *Check modeling style for component deployment*
Not Run

 *Check signal interfaces*
Not Run

 *Check configuration for component deployment using service interface configuration*
Not Run

 *Check startup and shutdown event*
Not Run

 *Check usage of partial data send*
Not Run

 Modeling Standards for DO-178C/DO-331 0 0 0 0 0 0 103

 *Display model version information*
Not Run

 High-Integrity Systems 0 0 0 0 0 0 96

 Simulink 0 0 0 0 0 0 38

 *Check usage of Abs blocks*
Not Run

 *Check usage of remainder and reciprocal operations*
Not Run

 *Check usage of square root operations*
Not Run

 Check usage of log and log10 operations
Not Run

 Check usage of While Iterator blocks
Not Run

 Check usage of For and While Iterator subsystems
Not Run

 Check usage of For Iterator blocks
Not Run

 Check usage of If blocks and If Action Subsystem blocks
Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check usage of Merge blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of variant blocks
Not Run

 Check usage of lookup table blocks
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for unreachable and dead code
Not Run

 Check for invalid root input and output port connections
Not Run

 Stateflow 0 0 0 0 0 0 11

 Check state machine type of Stateflow charts
Not Run

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 MATLAB 0 0 0 0 0 0  11

 Check usage of standardized MATLAB function headers
Not Run

 Check for MATLAB Function interfaces with inherited properties
Not Run

 Check MATLAB Function metrics
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Metrics for generated code complexity
Not Run

 Configuration 0 0 0 0 0 0 31

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 0 2

 Check model file name
Not Run

 Check model element names
Not Run

 Requirements 0 0 0 0 0 0 1

 Check for model elements that do not link to requirements
Not Run

 Code  0  0  0  0  0  0  2

 Check for blocks not recommended for MISRA C:2012
Not Run

 Check configuration parameters for MISRA C:2012
Not Run

 *Simulink*  0  0  0  0  0  0  1

 Identify unconnected lines, input ports, and output ports
Not Run

 *Library Links*  0  0  0  0  0  0  1

 Identify unresolved library links
Not Run

 *Requirements Consistency*  0  0  0  0  0  0  4

 Identify requirement links that specify invalid locations within documents
Not Run

 Identify requirement links with missing documents
Not Run

 Identify requirement links with path type inconsistent with preferences
Not Run

 Identify selection-based links having description fields that do not match their requirements document text
Not Run

 Modeling Standards for DO-254 0 0 0 0 0 93

 *Display model version information*

Not Run

 High-Integrity Systems 0 0 0 0 0 51

 Simulink 0 0 0 0 0 23

 Check usage of Abs blocks

Not Run

 Check relational comparisons on floating-point signals

Not Run

 Check usage of Relational Operator blocks

Not Run

 Check usage of Logical Operator blocks

Not Run

 Check usage of bitwise operations

Not Run

 Check for inconsistent vector indexing methods

Not Run

 Check data types for blocks with index signals

Not Run

 Check for root Input ports with missing properties

Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for unreachable and dead code
Not Run

 Check for invalid root input and output port connections
Not Run

 Stateflow 0 0 0 0 0 0 8

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check Stateflow charts for unary operators
Not Run

 MATLAB  0  0  0  0  0  0  7

 Check usage of standardized MATLAB function headers
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 0 2

 Check model file name
Not Run

 Check model element names
Not Run

 Requirements 0 0 0 0 0 0 1

 Check for model elements that do not link to requirements
Not Run

 Library Links 0 0 0 0 0 0 1

 Identify unresolved library links
Not Run

 Requirements Consistency 0 0 0 0 0 0 4

 Identify requirement links that specify invalid locations within documents
Not Run

 Identify requirement links with missing documents
Not Run

 Identify requirement links with path type inconsistent with preferences
Not Run

 Identify selection-based links having description fields that do not match their requirements
document text
Not Run

 HDL Coder 0 0 0 0 0 36

 Checks for blocks and block settings 0 0 0 0 0 10

 Check for HDL Reciprocal block usage
Not Run

 Check for infinite and continuous sample time sources
Not Run

 Check for unsupported blocks
Not Run

 Check for MATLAB Function block settings
Not Run

 Check for Stateflow chart settings
Not Run

 Check for Trigonometric Function block for LUT-based approximation method
Not Run

 Check for obsolete Unit Delay Enabled/Resettable blocks
Not Run

 Check for unsupported storage class for signal objects
Not Run

 Check for large matrix operations
Not Run

 Identify unconnected lines, input ports, and output ports
Not Run

 Industry standard checks 0 0 0 0 0 11

 Check file extension
Not Run

 Check naming conventions
Not Run

 Check top-level subsystem/port names
Not Run

 Check module/entity names
Not Run

 Check package file names
Not Run

 Check signal and port names
Not Run

 Check generics
Not Run

 Check clock, reset, and enable signals
Not Run

 Check architecture name
Not Run

 Check entity and architecture
Not Run

 Check clock settings
Not Run

 Model configuration checks 0 0 0 0 0 0 6

 Check for model parameters suited for HDL code generation
Not Run

 Check for global reset setting for Xilinx and Altera devices
Not Run

 Check inline configurations setting
Not Run

 Check for visualization settings
Not Run

 Check delay balancing setting
Not Run

 Check algebraic loops
Not Run

 Native Floating Point checks 0 0 0 0 0 0 8

 Check for blocks that have nonzero output latency
Not Run

 Check blocks with nonzero ulp error
Not Run

 Check for single datatypes in the model
Not Run

 Check for double datatypes in the model
Not Run

 Check for Data Type Conversion blocks with incompatible settings
Not Run

 Check for HDL Reciprocal block usage
Not Run

 Check for Relational Operator block usage
Not Run

 Check for unsupported blocks with Native Floating Point
Not Run

 Checks for ports and subsystems 0 0 0 0 0 1

 Check for invalid top level subsystem
Not Run

 Modeling Standards for IEC 61508 0 0 0 0 0 99

 *Display configuration management data*
Not Run

 *Display model metrics and complexity report*
Not Run

 *Check for unconnected objects*
Not Run

 *High-Integrity Systems* 0 0 0 0 0 0 96

 *Simulink* 0 0 0 0 0 0 38

 *Check usage of Abs blocks*
Not Run

 *Check usage of remainder and reciprocal operations*
Not Run

 *Check usage of square root operations*
Not Run

 *Check usage of log and log10 operations*
Not Run

 *Check usage of While Iterator blocks*
Not Run

 *Check usage of For and While Iterator subsystems*
Not Run

 *Check usage of For Iterator blocks*
Not Run

 Check usage of If blocks and If Action Subsystem blocks
Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check usage of Merge blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of variant blocks
Not Run

 Check usage of lookup table blocks
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for unreachable and dead code
Not Run

 Check for invalid root input and output port connections
Not Run

 Stateflow 0 0 0 0 0 0 11

 Check state machine type of Stateflow charts
Not Run

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 Check usage of standardized MATLAB function headers

Not Run

 Check for MATLAB Function interfaces with inherited properties

Not Run

 Check MATLAB Function metrics

Not Run

 Check MATLAB Code Analyzer messages

Not Run

 Check if/elseif/else patterns in MATLAB Function blocks

Not Run

 Check switch statements in MATLAB Function blocks

Not Run

 Check usage of relational operators in MATLAB Function blocks

Not Run

 Check usage of logical operators and functions in MATLAB Function blocks

Not Run

 Check type and size of condition expressions

Not Run

 Check MATLAB functions not supported for code generation

Not Run

 Metrics for generated code complexity
Not Run

 Configuration  0  0  0  0  0  0  31

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 2

 Check model file name
Not Run

 Check model element names
Not Run

 Requirements 0 0 0 0 0 1

 Check for model elements that do not link to requirements
Not Run

 Code 0 0 0 0 0 2

 Check for blocks not recommended for MISRA C:2012
Not Run

 Check configuration parameters for MISRA C:2012
Not Run

 Modeling Standards for IEC 62304 0 0 0 0 0 99

 *Display configuration management data*
Not Run

 *Display model metrics and complexity report*
Not Run

 Check for unconnected objects

Not Run

 High-Integrity Systems 0 0 0 0 0 0 96

 Simulink 0 0 0 0 0 0 38

 Check usage of Abs blocks

Not Run

 Check usage of remainder and reciprocal operations

Not Run

 Check usage of square root operations

Not Run

 Check usage of log and log10 operations

Not Run

 Check usage of While Iterator blocks

Not Run

 Check usage of For and While Iterator subsystems

Not Run

 Check usage of For Iterator blocks

Not Run

 Check usage of If blocks and If Action Subsystem blocks

Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check usage of Merge blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of variant blocks
Not Run

 Check usage of lookup table blocks
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for unreachable and dead code
Not Run

 Check for invalid root input and output port connections
Not Run

 Stateflow 0 0 0 0 0 0 11

 Check state machine type of Stateflow charts
Not Run

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 MATLAB  0  0  0  0  0  11

 Check usage of standardized MATLAB function headers
Not Run

 Check for MATLAB Function interfaces with inherited properties
Not Run

 Check MATLAB Function metrics
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Metrics for generated code complexity
Not Run

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 2

 Check model file name

Not Run

 Check model element names

Not Run

 Requirements 0 0 0 0 0 1

 Check for model elements that do not link to requirements

Not Run

 Code 0 0 0 0 0 2

 Check for blocks not recommended for MISRA C:2012

Not Run

 Check configuration parameters for MISRA C:2012

Not Run

 Modeling Standards for ISO 26262 1 7 26 0 65 0

 *Display configuration management data*

Display model configuration and checksum information

Model configuration and checksum information

Attribute	Value
Model Version	11.39

Author	The MathWorks, Inc.
Date	Tue Apr 28 21:20:44 2026
Model Checksum	2460031036 1930952176 372521297 1912146345

✔ *Display model metrics and complexity report*

Display number of elements and name, level, and depth of subsystems for the model or subsystem

Model metrics information

Display number of elements for Simulink blocks and Stateflow constructs

Summary

Element Type	Count
Inport	12
Outport	10
SubSystem	7

Simulink

Block Type	Count
Inport	12
Outport	10
SubSystem	7
Gain	6

Lookup_n-D	4
Constant	3
ModelReference	2
Scope	2
ZeroOrderHold	2
Sum	1
Switch	1
BusCreator	1
RateTransition	1
Fcn	1
UnitDelay	1
MultiPortSwitch	1
Step	1
Stop	1
Integrator	1
DiscreteIntegrator	1

Λ Less

Model complexity information

Display name, level, and depth of subsystems

Maximum Subsystem Depth: 3

Subsystem Depth

Subsystem Name	Level	Depth
sldemo_absbrake/Compare To Constant	1	1
sldemo_absbrake/More Info	1	1
sldemo_absbrake/S-R Flip-Flop	1	1
sldemo_absbrake/Vehicle Dynamics	1	2
sldemo_absbrake/Vehicle Dynamics/Relative Slip	2	1
sldemo_absbrake/Vehicle Dynamics/Vehicle	2	1
sldemo_absbrake/Visualization	1	1

Check for unconnected objects

Identify unconnected lines, input ports, and output ports in the model

Warning

The following lines, input ports, or output ports are not properly connected in system:
sldemo_absbrake

- sldemo_absbrake/S-R Flip-Flop

Recommended Action

Connect the blocks specified in the list

 High-Integrity Systems  1  7  25  0  63  0  0

 Simulink  0  6  4  0  28  0  0

✔ Check usage of Abs blocks

Identify Abs blocks that have unreachable code or produce overflows.

Passed

No Abs blocks found causing unreachable code or produce overflows.

✘ Check usage of remainder and reciprocal operations

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianness' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

✘ Check usage of square root operations

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianness' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

✘ Check usage of log and log10 operations

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianness' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

✔ Check usage of While Iterator blocks

Identify While Iterator blocks that do not have a positive value for the maximum number of iterations.

Passed

No While Iterator blocks found that might cause infinite loops

- ✓ Check usage of For and While Iterator subsystems
Identify sample time-dependent blocks in While and For Iterator subsystems.

Passed

No sample time-dependent blocks in For or While Iterator subsystems.

- ✓ Check usage of For Iterator blocks
Identify For Iterator blocks that cause variable loops.

Passed

No For Iterator blocks found that cause variable loops.

- ✓ Check usage of If blocks and If Action Subsystem blocks
Identify If and If Action Subsystem blocks without else conditions.

Passed

No If blocks with questionable configurations or connections were found.

- ✓ Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Identify inappropriately used Switch Case blocks and Switch Case Action Subsystem blocks.

Passed

No Switch Case blocks with questionable configurations or connections were found.

- ✓ Check usage of Merge blocks
Identify Merge blocks constructs which can lead to ambiguous behavior.

Passed

No merge blocks found which can lead to ambiguous behavior.

- ✓ Check relational comparisons on floating-point signals
Identify relational blocks or operations that perform equality or inequality comparisons on floating-point data types.

Passed

No relational blocks or operations found that perform equality or inequality comparisons on floating-point data types.

✓ Check usage of Relational Operator blocks

Identify Relational Operator blocks that operate on different data types or have a non-boolean output.

Passed

No Relational Operator blocks found that operate on different data types or have a non-boolean output.

✓ Check usage of Logical Operator blocks

Identify Logical Operator blocks that operate on non-boolean data types.

Passed

No Logical Operator blocks found that operate on non-boolean data types.

✓ Check usage of bitwise operations

Identify bit operations with signed data types as inputs.

Passed

No bit operation found with signed data types as inputs.

⚠ Check for blocks not recommended for C/C++ production code deployment

Identify blocks not supported by code generation or not recommended for C/C++ production code deployment.

Warning

The following blocks are not recommended for C/C++ production code generation:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Step
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Stop Simulation2

Recommended Action

Although Embedded Coder supports these blocks, they are not recommended for C/C++ production code deployment.

blocks not supported by code generation or not recommended for C/C++ production code deployment.

Warning

The following blocks are not recommended for C/C++ production code generation:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Stopping distance

Recommended Action

Although Embedded Coder supports these blocks, they are not recommended for C/C++ production code deployment.

1. Consider using the Model Discretizer to map these continuous blocks into discrete equivalents that support code generation. From a model, select Model Discretizer to access the Model Discretizer.

2. Not recommended for production code.

Check for inconsistent vector indexing methods

Identify inconsistent usage of vector indexing methods across the model or subsystem.

Passed

No blocks found using inconsistent indexing modes.

Check data types for blocks with index signals

Identify blocks with index signals that have data types other than integer or enum.

Warning

The following blocks have inappropriate data types for index signals or variables:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Multiport Switch

Recommended Action

Change the data type of index signals or variables to an integer or enum data type.

Check usage of variant blocks

Check variant block settings that might result in code that doesn't trace back to requirements.

Passed

No variant blocks have "VariantActivationTime" set to 'code compile'.

Check usage of lookup table blocks

Check for Lookup Table blocks, Prelookup blocks and Interpolation blocks that do not generate out-of-range checking code.

Passed

No lookup table blocks found to not generate out-of-range checking code.

Check usage of Signal Routing blocks

Identify usage of Signal Routing blocks in Simulink that might impact safety.

Passed

No Switch blocks that might generate code with inequality operations ($\sim=$) in expressions where at least one side of the expression is a floating-point variable or constant were found.

Check for root Input ports with missing properties

Passed

Check for root Input ports with missing range definitions

Passed

Check for root Output ports with missing range definitions

Passed

 Check usage of Reciprocal Sqrt blocks

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianess' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

 Check usage of Assignment blocks

Identify Assignment blocks whose array fields are not initialized.

Passed

All Assignment blocks are configured with block parameter "Action if any output element is not assigned" set to Error.

 Check global variables in graphical functions

Identify expressions that both read and write to the same global data.

Passed

No expressions found that both read and write to the same global data.

 Check usage of Gain blocks

Identify Gain blocks with value which resolves to 1.

Warning

The following Gain blocks have value which resolves to 1.

- sldemo_absbrake/ctrl

Recommended Action

Consider remodeling to remove the Gain blocks with values that resolve to 1

✓ Check for length of user-defined object names

Identify user-defined object names with length greater than threshold.

Passed

No Subsystem blocks found with function name length greater than Maximum identifier length. There are no data objects with names having length greater than Maximum identifier length.

✓ Check data type of loop control variables

Identify loop control variables using non-integer data types.

Passed

No For Iterator blocks or MATLAB Function blocks found using non-integer data type for loop control counter variable.

✗ Check for divide-by-zero calculations

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianness' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

✓ Check for parameter tunability ignored for referenced models

Check for models parameter tunability information specified using Model Parameter Configuration dialog boxes.

Passed

No parameters found that lose the tunability defined in the referenced models.

✓ Check usage of bit-shift operations

Identify blocks or expressions that perform bit-shift operations greater than the bit width of Input type.

Passed

No blocks or expressions found that perform bit-shift operations greater than the bit width of Input type.

⚠ Check safety-related diagnostic settings for variants

Check diagnostic settings in the model configuration that apply to variants and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Arithmetic operations in variant conditions (ArithmeticOperatorsInVariantConditions)	warning	error
Warning	Variant condition mismatch at signal source and destination (VariantConditionMismatch)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

✔ Check for disabled and parameterized library links

Identify disabled and parameterized library links in the model.

Passed

No blocks found that have disabled or parameterized library links.

✔ Check for root Outports with missing properties

Passed

✔ Check usage of identical modeling patterns

Identify identical modeling patterns that can increase the complexity of model and generated code.

Passed

No usage of identical modeling patterns are found.

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianess' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

Detect invalid root input and output port connections in referenced models.

Passed

Root input and output port connections are valid.

Stateflow 0 1 1 0 9 0 0

Identify Stateflow Charts whose State Machine Type differs from the type set in the Model Advisor Configuration Editor.

Passed

No Stateflow Charts found that deviate from recommended state machine type.

Error occurred during model compile.**NOTE:** This check applies to models configured for code generation.Build method inconsistency between top model sldemo_absbrake (Toolchain) and referenced model sldemo_wheelspeed_absbrake (Template makefile). Models in the model reference hierarchy must all use either the toolchain or template makefile approach.Model sldemo_wheelspeed_absbrake uses the template makefile build approach because one of the following settings is not a default value: TemplateMakefile: "rsim_default_tmf"
 MakeCommand: "make_rtw" RTWCompilerOptimization: "off"
 UseToolchainInfoCompliant: "off"

Identify whether Stateflow debugging options are set appropriately.

Warning

The following Stateflow debugging options are not set appropriately:

Parameter	Current Value	Recommended Values
Simulation range checking (SignalRangeChecking)	none	error

Recommended Action

Change the Stateflow debugging options to the recommended value.

-
- ✔ Check Stateflow charts for transition paths that cross parallel state boundaries
Identify transition paths that cross parallel state boundaries in Stateflow charts.

Passed

No transition paths crossing parallel state boundaries were found in Stateflow charts.

-
- ✔ Check for inappropriate use of transition paths
Identify transition paths that go into and out of a state without ending on a substate.

Passed

No transition paths found that go into and out of a state without ending on a substate.

-
- ✔ Check Stateflow charts for strong data typing
Identify expressions with variables and parameters of different data types in Stateflow objects.

Passed

No expressions were found with variables and parameters of different data types.

✔ Check naming of ports in Stateflow charts

Identify mismatches between names of Stateflow ports and associated signals.

Passed

There are no name mismatches between Stateflow ports and associated signals

✔ Check scoping of Stateflow data objects

Identify Stateflow data objects with local scope that are not scoped at the chart level or below.

Passed

All Stateflow data objects are properly scoped.

✔ Check Stateflow charts for uniquely defined data objects

Identify local data identifiers that are defined in multiple scopes within a chart.

Passed

No Stateflow data identifiers found to be defined in multiple scopes.

✔ Check assignment operations in Stateflow charts

Identify assignment operations in Stateflow objects which cast integer and fixed-point calculations to wider datatype.

Passed

No assignment operations were found which cast integer and fixed-point calculations to wider datatype.

✔ Check Stateflow charts for unary operators

Identify unary minus operators on unsigned data types in Stateflow objects.

Passed

No unary minus operations on unsigned data types were found in Stateflow objects.

 MATLAB 0 0 0 0 11 0 0

✔ Check usage of standardized MATLAB function headers

Identify usage of standardized function headers in MATLAB function.

Passed

No MATLAB function blocks found without standardized function headers.

✔ Check for MATLAB Function interfaces with inherited properties

Identify MATLAB Functions that have inputs, outputs, or parameters with inherited complexity or data type properties.

Passed

No MATLAB Function interfaces with inherited complexity or data type properties found.

✔ Check MATLAB Function metrics

Identify MATLAB Functions that violate code and complexity metrics.

Passed

No MATLAB Function blocks found that violate code and complexity metrics.

✔ Check MATLAB Code Analyzer messages

Check MATLAB functions for %#codegen directive, MATLAB Code Analyzer messages, and justification message IDs.

Passed

No MATLAB Function blocks found with Code Analyzer messages, missing %#codegen directive or inappropriate usage of justification message IDs.

✔ Check if/elseif/else patterns in MATLAB Function blocks

Identify if/elseif/else patterns without appropriate else conditions in embedded MATLAB code.

Passed

No inappropriate if/elseif/else patterns found.

✔ Check switch statements in MATLAB Function blocks

Identify inappropriately used switch statements in embedded MATLAB code.

Passed

No inappropriately used switch statements found.

- ✓ Check usage of relational operators in MATLAB Function blocks
Identify relational operators operating on operands of different data types in MATLAB Function blocks.

Passed

No relational operators found operating on operands of different data types.

- ✓ Check usage of logical operators and functions in MATLAB Function blocks
Identify logical operators and functions operating on operands with numerical data types.

Passed

No logical operators or functions found operating on operands with numerical data types.

- ✓ Check type and size of condition expressions
Identify condition expressions which are not logical scalars.

Passed

No condition expressions found which are not logical scalars.

- ✓ Check MATLAB functions not supported for code generation
Identify MATLAB functions that are not supported for code generation.

Passed

All identified MATLAB functions are supported for code generation.

- ✓ Metrics for generated code complexity
Identify usage of built-in MATLAB Functions with high code complexity.

Passed

No MATLAB constructs found with high code complexity.

 Configuration  0  0  19  0  12  0  0

- ⚠ Check safety-related diagnostic settings for data store memory
Check diagnostic settings in the model configuration that apply to data store memory and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Detect read before write (ReadBeforeWriteMsg)	UseLocalSettings	EnableAllAsError
Warning	Detect write after read (WriteAfterReadMsg)	UseLocalSettings	EnableAllAsError
Warning	Detect write after write (WriteAfterWriteMsg)	UseLocalSettings	EnableAllAsError

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 [Check safety-related diagnostic settings for saving](#)

Check diagnostic settings in the model configuration that apply to saving model files.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Block diagram contains disabled library links (SaveWithDisabledLinksMsg)	warning	error
Warning	Block diagram contains parameterized library links (SaveWithParameterizedLinksMsg)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related model referencing settings](#)

Check model referencing settings in the model configuration that might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Rebuild (UpdateModelReferenceTargets)	IfOutOfDate	AssumeUpToDate, IfOutOfDateOrStructuralChange

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 [Check safety-related code generation settings for comments](#)

Check code generation settings in the model configuration that apply comments and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values	Prerequisites
Warning	Requirements in block comments (ReqsInCode)	off	on	SystemTargetFile, GenerateComments
Warning	Stateflow object comments (StateflowObjectComments)	off	on	GenerateComments
Warning	MATLAB source code as comments (MATLABSourceComments)	off	on	GenerateComments

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 [Check safety-related code generation interface settings](#)

Check code generation interface settings in the model configuration that might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values	Prerequisites
Warning	non-finite numbers (SupportNonFinite)	on	off	
Warning	Terminate function required (IncludeMdlTerminateFcn)	on	off	SystemTargetFile
Warning	Remove error status field in real-time model data structure (SuppressErrorStatus)	off	on	SystemTargetFile

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 [Check safety-related solver settings for simulation time](#)

Identify if the model Start time is set to 0 and Stop time is less than the Application Life Span.

Passed

No issues found with solver settings for simulation time.

 [Check safety-related solver settings for solver options](#)

Check solver settings in the model configuration that apply to solvers and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Type (SolverType)	Variable-step	Fixed-step
Warning	Solver (SolverName)	ode45	FixedStepDiscrete

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

Check safety-related solver settings for tasking and sample-time

Check solver settings in the model configuration that apply to tasking and sample-time constraints and might impact safety.

Passed

Check safety-related diagnostic settings for solvers

Check diagnostic settings in the model configuration that apply to solvers and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Algebraic loop (AlgebraicLoopMsg)	warning	error
Warning	Artificial algebraic loop occurrences not eliminated (ArtificialAlgebraicLoopMsg)	warning	error

Warning	Block priority violation (BlockPriorityViolationMsg)	warning	error
Warning	Automatic solver parameter selection (SolverPrmCheckMsg)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related diagnostic settings for sample time](#)

Check diagnostic settings in the model configuration that apply to sample time and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Source block specifies -1 sample time (InheritedTsInSrcMsg)	warning	error
Warning	Enforce sample times specified by Signal Specification blocks (SigSpecEnsureSampleTimeMsg)	none	error
Warning	Tasks with equal priority (TasksWithSamePriorityMsg)	warning	error
Warning	Unspecified inheritability of sample time (UnknownTsInhSupMsg)	warning	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related optimization settings for logic signals](#)

Check optimization settings in the model configuration that apply to logic signals and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Implement logic signals as Boolean data (vs. double) (BooleanDataType)	on	on

[Check safety-related code generation settings for code style](#)

Check code generation settings in the model configuration that apply to code style and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values	Prerequisites
Warning	Preserve operand order in expression (PreserveExpressionOrder)	off	on	SystemTargetFile

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related optimization settings for application lifespan](#)

Check optimization settings in the model configuration that apply to application lifespan and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Application lifespan (days) (LifeSpan)	inf	Inf

[Check safety-related code generation identifier settings](#)

Check code generation identifier settings in the model configuration that might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Not Recommended Values	Prerequisites
Warning	Minimum mangle length (MangleLength)	1	1, 2, 3	SystemTargetFile

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related optimization settings for data initialization](#)

Check optimization settings in the model configuration that apply to data initialization and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values	Prerequisites
Warning	Remove root level I/O zero initialization (ZeroExternalMemoryAtStartup) *	off	on	SystemTargetFile, CodeInterfacePackaging

Warning	Remove internal data zero initialization (ZeroInternalMemoryAtStartup) *	off	on	SystemTargetFile, CodeInterfacePackaging
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Recommended Action

Follow the links in the result table to modify the model configuration parameters.

* The Command-Line values provided in the table are reverse of the settings in the Configuration Parameters Dialog. Therefore, 'on' in the Command-Line corresponds to an "Off" setting in the dialog, and 'off' in the Command-Line corresponds to an "On" setting in the dialog.

Check [safety-related optimization settings for data type conversions](#)

Check optimization settings in the model configuration that apply to data type conversions and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Remove code from floating-point to integer conversions that wraps out-of-range values (EfficientFloat2IntCast)	on	on

✔ Check safety-related optimization settings for division arithmetic exceptions

Check optimization settings in the model configuration that apply to division arithmetic exceptions and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values	Prerequisites
D - Pass	System target file (SystemTargetFile)	ERT based target	ERT based target	
Pass	Remove code that protects against division arithmetic exceptions (NoFixptDivByZeroProtection)	off	off	SystemTargetFile

✔ Check safety-related optimization settings for specified minimum and maximum values

Check optimization settings in the model configuration that apply to specified minimum and maximum values and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values	Prerequisites
--------	-----------	---------------	--------------------	---------------

Pass	Optimize using the specified minimum and maximum values (UseSpecifiedMinMax)	off	off	SystemTargetFile
D - Pass	System target file (SystemTargetFile)	ERT based target	ERT based target	

✓ Check safety-related settings for hardware implementation

Identify inconsistencies or underspecification of hardware attributes that can lead to incorrect and inefficient generated code.

Check 'Byte ordering' and 'Signed integer division rounds to' parameters

Passed

Target specification is consistent.

Search for 'Test hardware is the same as production hardware' in the Configuration Parameters dialog box and check if it is selected. If it is cleared, identify whether target specifications match.

Check whether 'Production hardware' and 'Test hardware' match

Passed

'Test hardware is the same as production hardware' is selected or is cleared and the target specifications match.

⚠ Check safety-related diagnostic settings for compatibility

Check diagnostic settings in the model configuration that affect compatibility and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	S-function upgrades needed (SFcnCompatibilityMsg)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

Check safety-related diagnostic settings for parameters

Check diagnostic settings in the model configuration that apply to parameters and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Detect underflow (ParameterUnderflowMsg)	none	error
Warning	Detect precision loss (ParameterPrecisionLossMsg)	warning	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

✔ Check safety-related diagnostic settings for Merge blocks

Check diagnostic settings in the model configuration that apply to Merge blocks and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Detect multiple driving blocks executing at the same time step (MergeDetectMultiDrivingBlocksExec)	error	error

✔ Check safety-related diagnostic settings for model initialization

Check diagnostic settings in the model configuration that affect model initialization and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Underspecified initialization detection (UnderspecifiedInitializationDetection)	Simplified	Simplified

✔ Check safety-related diagnostic settings for data used for debugging

Check diagnostic settings in the model configuration that apply to data used for debugging and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Model Verification block enabling (AssertControl)	DisableAll	DisableAll

⚠ Check safety-related diagnostic settings for signal connectivity

Check diagnostic settings in the model configuration that apply to signal connectivity and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
--------	-----------	---------------	--------------------

Warning	Signal label mismatch (SignalLabelMismatchMsg)	none	error
Warning	Unconnected block input ports (UnconnectedInputMsg)	warning	error
Warning	Unconnected block output ports (UnconnectedOutputMsg)	warning	error
Warning	Unconnected line (UnconnectedLineMsg)	warning	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related diagnostic settings for bus connectivity](#)

Check diagnostic settings in the model configuration that apply to bus connectivity and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Unspecified bus object at root Output block (RootOutputRequireBusObject)	warning	error
Warning	Element name mismatch (BusObjectLabelMismatch)	none	error

Warning	Bus signal treated as vector (StrictBusMsg)	ErrorLevel1	ErrorOnBusTreatedAsVector
Warning	Non-bus signals treated as bus signals (NonBusSignalsTreatedAsBus)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

-
- ✔ [Check safety-related diagnostic settings that apply to function-call connectivity](#)
Check diagnostic settings in the model configuration that apply to function-call connectivity and might impact safety.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	InvalidFcnCallConnMsg	error	error
Pass	Context-dependent inputs (FcnCallInpInsideContextMsg)	error	error

-
- ⚠ [Check safety-related diagnostic settings for type conversions](#)
Check diagnostic settings in the model configuration that apply to type conversions and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Unnecessary type conversions (UnnecessaryDatatypeConvMsg)	none	warning
Warning	Vector/matrix block input conversion (VectorMatrixConversionMsg)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related diagnostic settings for model referencing](#)

Check diagnostic settings in the model configuration that apply to model referencing and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
--------	-----------	---------------	--------------------

Warning	Port and parameter mismatch (ModelReferenceIOMismatchMessage)	none	error
Warning	Unsupported data logging (ModelReferenceDataLoggingMessage)	warning	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related diagnostic settings for Stateflow](#)

Check diagnostic settings in the model configuration that apply to Stateflow and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Unexpected backtracking (SFUnexpectedBacktrackingDiag)	warning	error
Warning	Invalid input data access in chart initialization (SFInvalidInputDataAccessInChartInitDiag)	warning	error
Warning	No unconditional default transitions (SFNoUnconditionalDefaultTransitionDiag)	warning	error
Warning	Transition outside natural parent (SFTransitionOutsideNaturalParentDiag)	warning	error

Warning	Unreachable execution path (SFUnreachableExecutionPathDiag)	warning	error
Warning	Transition action specified before condition action (SFTransitionActionBeforeConditionDiag)	warning	error
Warning	Absolute time temporal value shorter than sampling period (SFTemporalDelaySmallerThanSampleTimeDiag)	warning	error
Warning	Self-transition on leaf state (SFSelfTransitionDiag)	warning	error
Warning	'Execute-at-initialization' disabled in presence of input events (SFExecutionAtInitializationDiag)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

[Check safety-related diagnostic settings for signal data](#)

Check diagnostic settings in the model configuration that apply to signal data and might impact safety.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Division by singular matrix (CheckMatrixSingularityMsg)	none	error

Warning	Underspecified data types (UnderSpecifiedDataTypeMsg)	none	error
Warning	Saturate on overflow (IntegerSaturationMsg)	none	error
Warning	Inf or NaN block output (SignalInfNanChecking)	warning	error
Warning	Simulation range checking (SignalRangeChecking)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 Naming 0 0 1 0 1 0 0

Check model file name

Identify inappropriate characters and length issues in model file name.

Passed

No issues found with model file name.

Check model element names

Check Signal names

Identify invalid names of Signals

Warning

Contains prohibited characters

Signal	Issue
vehicle deceleration	vehicledeceleration
Tire Torque	TireTorque
Vehicle Velocity	VehicleVelocity
Vehicle Velocity	VehicleVelocity
Vehicle Velocity	VehicleVelocity

Recommended Action

Change flagged names of signals

Check

Block names

Identify invalid names of Blocks

Warning

Contains prohibited characters

Block Path	Expression
sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle speed (angular)	Vehiclespeed(angular)
sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle Speed	VehicleSpeed
sldemo_absbrake/Vehicle Dynamics/Vehicle/-1//m	-1/m

sldemo_absbrake/Vehicle Dynamics/Unit Delay	UnitDelay
sldemo_absbrake/Zero-Order Hold	Zero-OrderHold
sldemo_absbrake/More Info	MoreInfo
sldemo_absbrake/Vehicle Dynamics	VehicleDynamics
sldemo_absbrake/Vehicle Dynamics/Wheel Speed	WheelSpeed
sldemo_absbrake/Vehicle Dynamics/Vehicle/Tire Torque	TireTorque
sldemo_absbrake/Vehicle Dynamics/Controller out	Controllerout
sldemo_absbrake/Always Latch	AlwaysLatch
sldemo_absbrake/Compare To Constant	CompareToConstant
sldemo_absbrake/Target Slip	TargetSlip
sldemo_absbrake/Ice Mode	IceMode
sldemo_absbrake/Vehicle Dynamics/Vehicle/Stopping distance	Stoppingdistance
sldemo_absbrake/Vehicle Dynamics/Relative Slip/Rate Transition	RateTransition
sldemo_absbrake/Vehicle Dynamics/Vehicle/Multiport Switch	MultiportSwitch
sldemo_absbrake/Vehicle Dynamics/Relative Slip	RelativeSlip
sldemo_absbrake/Zero-Order Hold1	Zero-OrderHold1
sldemo_absbrake/Vehicle Dynamics/Relative Slip/Bus Creator	BusCreator
sldemo_absbrake/Vehicle Dynamics/Vehicle/Stop Simulation2	StopSimulation2
sldemo_absbrake/S-R Flip-Flop	S-RFlip-Flop
sldemo_absbrake/Dry Mode	DryMode
sldemo_absbrake/Vehicle Dynamics/Relative Slip/Relative Slip	RelativeSlip
sldemo_absbrake/Visualization/Braking cycles	Brakingcycles

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Recommended Action

Change flagged names of blocks

Check

Block names

Identify invalid names of Blocks

Warning

Contains prohibited characters, Starts with a number

Block Path	Expression
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table1	1-DLookupTable1
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table	1-DLookupTable
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table2	1-DLookupTable2

Recommended Action

Change flagged names of blocks

 Requirements 1 0 0 0 0 0 0

 Check for model elements that do not link to requirements

Abnormal exit: The block diagram 'simulink_extras' is not loaded.

✓ Check for blocks not recommended for MISRA C:2012

Passed

✓ Check configuration parameters for MISRA C:2012

Identify configuration parameters that might impact MISRA C:2012 compliant code generation.

Passed

All constraints on model configuration parameters have been met.

Stat us	Parameter	Current Value	Recommended Values	Prerequisites
Pass	Model Verification block enabling (AssertControl)	Disable All	DisableAll	
D - Pass	Shared code placement (UtilityFuncGeneration)	Shared location	Shared location	
Pass	Generate shared constants (GenerateSharedConstants)	off	off	UtilityFuncGener ation
D - Pass	System target file (SystemTargetFile)	ERT based target	ERT based target	
Pass	continuous time (SupportContinuousTime)	off	off	SystemTargetFile
Pass	Support non-inlined S-functions (SupportNonInlinedSFcns)	off	off	SystemTargetFile

Pass	MAT-file logging (MatFileLogging)	off	off	
Pass	Code replacement library (CodeReplacementLibrary)	None	None, AUTOSAR 4.0	
Pass	Parentheses level (ParenthesesLevel)	Standards	Standards, Maximum	SystemTargetFile
Pass	Suppress generation of default cases for switch statements if unreachable (SuppressUnreachableDefaultCases)	off	off	
Pass	Casting modes (CastingMode)	Standards	Standards	SystemTargetFile
Pass	System-generated identifiers (InternalIdentifier)	Shortened	Shortened	SystemTargetFile
Pass	Signed integer division rounds to (ProdIntDivRoundTo)	Zero	Zero, Floor	
Pass	Use division for fixed-point net slope computation (UseDivisionForNetSlopeComputation)	on	on, UseDivisionForReciprocalsOfIntegersOnly	
Pass	Replace multiplications by powers of two with signed bitwise shifts (EnableSignedLeftShifts)	off	off	SystemTargetFile
Pass	Allow right shifts on signed integers (EnableSignedRightShifts)	off	off	SystemTargetFile
Pass	Wrap on overflow (IntegerOverflowMsg)	error	warning, error	
Pass	Inf or NaN block output (SignalInfNanChecking)	warning	warning	

Pass	Dynamic memory allocation in MATLAB functions (MATLABDynamicMemAlloc)	off	off	
Pass	ExtMode	off	off	
Pass	Undirected event broadcasts (SFUndirectedBroadcastEventsDiag)	error	error	
Pass	Compile-time recursion limit for MATLAB functions (CompileTimeRecursionLimit)	0	0	
Pass	Enable run-time recursion for MATLAB functions (EnableRuntimeRecursion)	off	off	
D - Pass	Include comments (GenerateComments)	on	on	
Pass	MATLAB user comments (MATLABFcnDesc)	on	on	GenerateComments, SystemTargetFile
Pass	InstructionSetExtensions	None	None	

^ Less

 Modeling Standards for ISO 25119 0 0 0 0 0 0 99

 [Display configuration management data](#)

Not Run

 [Display model metrics and complexity report](#)

Not Run

 [Check for unconnected objects](#)

Not Run

 *High-Integrity Systems* 0 0 0 0 0  96

 *Simulink* 0 0 0 0 0  38

 Check usage of Abs blocks
Not Run

 Check usage of remainder and reciprocal operations
Not Run

 Check usage of square root operations
Not Run

 Check usage of log and log10 operations
Not Run

 Check usage of While Iterator blocks
Not Run

 Check usage of For and While Iterator subsystems
Not Run

 Check usage of For Iterator blocks
Not Run

 Check usage of If blocks and If Action Subsystem blocks
Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks
Not Run

 Check usage of Merge blocks
Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of variant blocks
Not Run

 Check usage of lookup table blocks
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for unreachable and dead code
Not Run

 Check for invalid root input and output port connections
Not Run

 Stateflow 0 0 0 0 0 0 11

 Check state machine type of Stateflow charts
Not Run

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 MATLAB 0 0 0 0 0 0 11

 Check usage of standardized MATLAB function headers
Not Run

 Check for MATLAB Function interfaces with inherited properties
Not Run

 Check MATLAB Function metrics
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Metrics for generated code complexity
Not Run

 Configuration 0 0 0 0 0 0 31

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 0 2

 Check model file name
Not Run

 Check model element names
Not Run

 Requirements 0 0 0 0 0 0 1

 Check for model elements that do not link to requirements
Not Run

 Code 0 0 0 0 0 0 2

 Check for blocks not recommended for MISRA C:2012
Not Run

 Check configuration parameters for MISRA C:2012
Not Run

 Modeling Standards for EN 50128/EN 50657 0 0 0 0 0 0 99

 *Display configuration management data*
Not Run

 *Display model metrics and complexity report*
Not Run

 *Check for unconnected objects*
Not Run

 High-Integrity Systems 0 0 0 0 0 0 96

 Check usage of Abs blocks

Not Run

 Check usage of remainder and reciprocal operations

Not Run

 Check usage of square root operations

Not Run

 Check usage of log and log10 operations

Not Run

 Check usage of While Iterator blocks

Not Run

 Check usage of For and While Iterator subsystems

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 Check usage of For Iterator blocks

Not Run

 Check usage of If blocks and If Action Subsystem blocks

Not Run

 Check usage of Switch Case blocks and Switch Case Action Subsystem blocks

Not Run

 Check usage of Merge blocks

Not Run

 Check relational comparisons on floating-point signals
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check usage of Logical Operator blocks
Not Run

 Check usage of bitwise operations
Not Run

 Check for blocks not recommended for C/C++ production code deployment
Not Run

 Check for inconsistent vector indexing methods
Not Run

 Check data types for blocks with index signals
Not Run

 Check usage of variant blocks
Not Run

 Check usage of lookup table blocks
Not Run

 Check usage of Signal Routing blocks
Not Run

 Check for root Input ports with missing properties
Not Run

 Check for root Input ports with missing range definitions
Not Run

 Check for root Output ports with missing range definitions
Not Run

 Check usage of Reciprocal Sqrt blocks
Not Run

 Check usage of Assignment blocks
Not Run

 Check global variables in graphical functions
Not Run

 Check usage of Gain blocks
Not Run

 Check for length of user-defined object names
Not Run

 Check data type of loop control variables
Not Run

 Check for divide-by-zero calculations
Not Run

 Check for parameter tunability ignored for referenced models
Not Run

 Check usage of bit-shift operations
Not Run

 Check safety-related diagnostic settings for variants
Not Run

 Check for disabled and parameterized library links
Not Run

 Check for root Outports with missing properties
Not Run

 Check usage of identical modeling patterns
Not Run

 Check for unreachable and dead code
Not Run

 Check for invalid root input and output port connections
Not Run

 Stateflow 0 0 0 0 0 11

 Check state machine type of Stateflow charts
Not Run

 Check usage of recursions
Not Run

 Check Stateflow debugging options
Not Run

 Check Stateflow charts for transition paths that cross parallel state boundaries
Not Run

 Check for inappropriate use of transition paths
Not Run

 Check Stateflow charts for strong data typing
Not Run

 Check naming of ports in Stateflow charts
Not Run

 Check scoping of Stateflow data objects
Not Run

 Check Stateflow charts for uniquely defined data objects
Not Run

 Check assignment operations in Stateflow charts
Not Run

 Check Stateflow charts for unary operators
Not Run

 MATLAB 0 0 0 0 0 0 11

 Check usage of standardized MATLAB function headers
Not Run

 Check for MATLAB Function interfaces with inherited properties
Not Run

 Check MATLAB Function metrics
Not Run

 Check MATLAB Code Analyzer messages
Not Run

 Check if/elseif/else patterns in MATLAB Function blocks
Not Run

 Check switch statements in MATLAB Function blocks
Not Run

 Check usage of relational operators in MATLAB Function blocks
Not Run

 Check usage of logical operators and functions in MATLAB Function blocks
Not Run

 Check type and size of condition expressions
Not Run

 Check MATLAB functions not supported for code generation
Not Run

 Metrics for generated code complexity
Not Run

 Configuration 0 0 0 0 0 0 31

 Check safety-related diagnostic settings for data store memory
Not Run

 Check safety-related diagnostic settings for saving
Not Run

 Check safety-related model referencing settings
Not Run

 Check safety-related code generation settings for comments
Not Run

 Check safety-related code generation interface settings
Not Run

 Check safety-related solver settings for simulation time
Not Run

 Check safety-related solver settings for solver options
Not Run

 Check safety-related solver settings for tasking and sample-time
Not Run

 Check safety-related diagnostic settings for solvers
Not Run

 Check safety-related diagnostic settings for sample time
Not Run

 Check safety-related optimization settings for logic signals
Not Run

 Check safety-related code generation settings for code style
Not Run

 Check safety-related optimization settings for application lifespan
Not Run

 Check safety-related code generation identifier settings
Not Run

 Check safety-related optimization settings for data initialization
Not Run

 Check safety-related optimization settings for data type conversions
Not Run

 Check safety-related optimization settings for division arithmetic exceptions
Not Run

 Check safety-related optimization settings for specified minimum and maximum values
Not Run

 Check safety-related settings for hardware implementation
Not Run

 Check safety-related diagnostic settings for compatibility
Not Run

 Check safety-related diagnostic settings for parameters
Not Run

 Check safety-related diagnostic settings for Merge blocks
Not Run

 Check safety-related diagnostic settings for model initialization
Not Run

 Check safety-related diagnostic settings for data used for debugging
Not Run

 Check safety-related diagnostic settings for signal connectivity
Not Run

 Check safety-related diagnostic settings for bus connectivity
Not Run

 Check safety-related diagnostic settings that apply to function-call connectivity
Not Run

 Check safety-related diagnostic settings for type conversions
Not Run

 Check safety-related diagnostic settings for model referencing
Not Run

 Check safety-related diagnostic settings for Stateflow
Not Run

 Check safety-related diagnostic settings for signal data
Not Run

 Naming 0 0 0 0 0 0 2

 Check model file name
Not Run

 Check model element names
Not Run

 Requirements 0 0 0 0 0 0 1

 Check for model elements that do not link to requirements
Not Run

 Code 0 0 0 0 0 0 2

 Check for blocks not recommended for MISRA C:2012
Not Run

 Check configuration parameters for MISRA C:2012
Not Run

 Model Metrics 0 0 0 0 0 0 10

 Count Metrics 0 0 0 0 0 0 7

 Simulink block metric
Not Run

 Subsystem metric
Not Run

 Library link metric
Not Run

 Effective lines of MATLAB code metric
Not Run

Stateflow chart objects metric
Not Run

Lines of code for Stateflow blocks metric
Not Run

Subsystem depth metric
Not Run

Complexity Metrics 0 0 0 0 0 0 1

Cyclomatic complexity metric
Not Run

Readability Metrics 0 0 0 0 0 0 2

Nondescriptive block name metric
Not Run

Data and structure layer separation metric
Not Run

Modeling Standards for MAB 0 1 38 0 121 0 0

Naming Conventions 0 0 5 0 23 0 0

General 0 0 1 0 3 0 0

Check file names (28-Apr-2026 21:52:27)
Identical file names on path

Warning

The following files have names which are identical to files present in MATLAB path:

File name	Identical file names
C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\ABS_Controller_core_psf.mexw64	C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\ABS_Controller_core_psf.mexw64 C:\Users\vipul\AppData\Roaming\MathWorks\MATLAB Add-Ons\Toolboxes\Arm_Cortex-M_Fast_Model_Support_Package\ABS_Controller_core_psf.mexw64

Recommended Action

Consider having unique file names.

Characters allowed for file names

Check the model file name to ensure that the name complies with the recommended guidelines.

Warning

The following files have invalid names:

- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\ABS_Controller_core_(modelrefpil)_cov.html
- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\Verification Report.mlx
- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\look1_binlx_c_(pil)_cov.html
- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\sldemo_absbrake.slx.autosave
- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\sldemo_absbrake.slx.original
- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\untitled.slx.autosave

Recommended Action

Consider having only alphanumeric characters and underscores in file name.

reserved MATLAB word **Single**

Check the model file name to ensure that the name complies with the recommended guidelines.

Warning

The following files have Reserved MATLAB words as the file name:

- C:\Users\vipul\Downloads\simulink_automotive_pil\simulink_automotive\ModelingAnAntiLockBrakingSystemExample\mu.mat

Recommended Action

Consider not having Reserved MATLAB word as the file name.

- ✓ Check folder names (28-Apr-2026 21:52:27)
Check the folder name to ensure that the name complies with the recommended guidelines.

Passed

All folders have correct names.

- ✓ Check length of model file name (28-Apr-2026 21:52:27)
Identify long model file names.

Passed

Model name is valid.

- ✓ Check length of folder name at every level of model path (28-Apr-2026 21:52:27)
Identify long folder names.

Passed

Folder names are valid.

 Content 0 0 4 0 20 0 0

- ✓ Check subsystem names (28-Apr-2026 21:52:27)
Passed

- ⚠ Check port block names (28-Apr-2026 21:52:27)
Characters allowed for port block names

Warning

The following port blocks have invalid names:

- sldemo_absbrake/Vehicle Dynamics/Controller out
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Tire Torque

Recommended Action

Consider having only alphanumeric characters and underscores in port block name.

 Check character usage in block names (28-Apr-2026 21:52:27)

Characters allowed for block names

Warning

The following blocks have invalid names:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle speed (angular)
- sldemo_absbrake/Target Slip
- sldemo_absbrake/Vehicle Dynamics/Vehicle/-1//m
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle Speed
- sldemo_absbrake/Dry Mode
- sldemo_absbrake/Ice Mode
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Stopping distance
- sldemo_absbrake/Vehicle Dynamics/Wheel Speed
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Relative Slip
- sldemo_absbrake/Always Latch
- sldemo_absbrake/Visualization/Braking cycles

^ Less

Recommended Action

Consider having only alphanumeric characters and underscores in block name.

reserved MATLAB word

Single

Warning

The following blocks have reserved MATLAB words as the block name:

- sldemo_absbrake/Visualization/Wheel

Recommended Action

Consider not having reserved MATLAB word as the block name.

✓ Check length of subsystem names (28-Apr-2026 21:52:27)
Passed

✓ Check length of block names (28-Apr-2026 21:52:27)
Passed

✓ Check length of Inport and Outport names (28-Apr-2026 21:52:27)
Passed

⚠ Check character usage in signal names and bus names (28-Apr-2026 21:52:27)
Characters allowed for signal names and bus names

Identify signal and bus names with invalid characters.

Warning

The following signals or buses have invalid names:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Tire Torque
- sldemo_absbrake/Vehicle Dynamics/Vehicle/vehicle _deceleration

Recommended Action

Consider having only alphanumeric characters and underscores in signal names and bus names.

-
- ✔ Check character usage in parameter names (28-Apr-2026 21:52:27)
Identify parameter names with invalid characters.

Passed

No invalid characters are used in parameter names.

-
- ✔ Check length of signal and bus names (28-Apr-2026 21:52:27)
Identify long signal and bus names.

Passed

All signal and bus names are valid.

-
- ✔ Check length of parameter names (28-Apr-2026 21:52:27)
Identify long parameter names.

Passed

All parameter names are valid.

-
- ✔ Check character usage in Stateflow data names (28-Apr-2026 21:52:27)
Identify Stateflow data names with invalid characters.

Passed

No invalid characters are used in Stateflow data names.

- ✓ Check length of Stateflow data name (28-Apr-2026 21:52:27)
Check if the length of Stateflow data names is within limit.

Passed

All Stateflow data names are valid.

- ✓ Check duplication of Simulink Data names (28-Apr-2026 21:52:27)
Simulink Data names should be unique across base workspace, model workspace and data dictionary.

Passed

All Simulink Data names are unique.

- ✓ Check unused data in Simulink Model (28-Apr-2026 21:52:27)
Identify unused data in model workspace and data dictionary.

Passed

No unused data found in model workspace and/or data dictionary.

- ✓ Check for unused data in Stateflow Charts (28-Apr-2026 21:52:27)
Checks if the model parameter 'Unused data, events, messages and functions' is not set to 'none'.

Passed

All constraints on model configuration parameters have been met.

Status	Parameter	Current Value	Recommended Values
Pass	Unused data, events, messages and functions (SFUnusedDataAndEventsDiag)	warning	error, warning

- ✔ Check usage of restricted variable names (28-Apr-2026 21:52:27)
Identify usage of reserved keywords in MATLAB Function blocks.

Passed

No variable names conflict with reserved keywords

- ✔ Check Bus and Enumeration data type names (28-Apr-2026 21:52:27)
Identify Bus and Enumeration data type names with invalid characters.

Passed

All the Bus and Enumeration data type names use valid characters.

- ✔ Check length of Bus and Enumeration data type names (28-Apr-2026 21:52:27)
Identify Bus and Enumeration data type names with long names.

Passed

All the Bus and Enumeration data type names are valid.

- ✔ Check data names in MATLAB Functions (28-Apr-2026 21:52:27)
Identify data names with invalid characters in MATLAB Functions.

Passed

All the data names in MATLAB Functions use valid characters.

- ✔ Check the length of data names in MATLAB Functions (28-Apr-2026 21:52:27)
Identify length of data names in MATLAB Functions.

Passed

All data name lengths in MATLAB Functions are under the acceptable limit.

- ⚠ Check for overlapping or intersecting blocks and signal lines (28-Apr-2026 21:52:27)
jc_0903_b1 : Check for signal lines overlapping or intersecting other signal lines or blocks

Identify blocks, block names, signal lines, signal labels, annotations, or block annotations that overlap or intersect visibly.

Warning

The following signal lines overlap or intersect:

- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake/slp
- sldemo_absbrake
- sldemo_absbrake

Recommended Action

Modify the Simulink model such that signal lines do not overlap or intersect.

jc_0903_a : Check for overlapping blocks, annotations, names, or labels

Identify blocks, block names, signal lines, signal labels, annotations, or block annotations that overlap or intersect visibly.

Warning

The following signal labels overlap signal lines, blocks, block names, annotations, or block annotations:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/vehicle _deceleration
- sldemo_absbrake/Visualization

Recommended Action

Modify the Simulink model such that signal labels do not overlap signal lines, blocks, block names, annotations, or block annotations.

-
- ✓ Check for overlapping of states and transition lines (28-Apr-2026 21:52:27)
Identify objects in Stateflow charts that visibly overlap.

Passed

Objects in Stateflow charts do not overlap.

-
- ✓ Check for arrowhead size of transition lines (28-Apr-2026 21:52:27)
Identify transition line arrowhead not set to size specified by the project setting.

Passed

All transition line arrowhead set to size specified by the project setting.

-
- ✓ Check for size of junctions (28-Apr-2026 21:52:27)
Identify junctions not set to size specified by the project setting.

Passed

All junctions set to size specified by the project setting.

📁 Simulink 0 1 31 0 31 0 0

📁 Configuration Parameters 0 0 2 0 2 0 0

-
- ✓ Check Implement logic signals as Boolean data (vs. double) (28-Apr-2026 21:52:27)
Identify whether **Implement logic signals as Boolean data (vs. double)** is selected.

Passed

Implement logic signals as Boolean data (vs. double) is selected.

-
- ✓ Check Signed Integer Division Rounding mode (28-Apr-2026 21:52:27)
Passed

 Check diagnostic settings for incorrect calculation results (28-Apr-2026 21:52:27)
Identify data validity diagnostic settings which detect incorrect calculation results.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Division by singular matrix (CheckMatrixSingularityMsg)	none	error
Warning	Inf or NaN block output (SignalInfNanChecking)	warning	error
Warning	Saturate on overflow (IntegerSaturationMsg)	none	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 Check model diagnostic parameters (28-Apr-2026 21:52:27)
Identify diagnostic parameters that are set to none.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Element name mismatch (BusObjectLabelMismatch)	none	error, warning

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

 Diagram Appearance  0  0  14  0  3  0  0

 [Check for Simulink diagrams using nonstandard display attributes \(28-Apr-2026 21:52:27\)](#)
Identify issues with model appearance.

Warning

The following diagrams do not have zoom factor set to 100 percent:

- sldemo_absbrake/Vehicle Dynamics

Recommended Action

For each listed diagram, select **Modeling > Environment > Zoom > Normal View (100%)**.

 [Check Model font settings \(28-Apr-2026 21:52:27\)](#)
Check font size in Simulink block and signal names

Identify blocks and charts with different font settings from input parameters.

Warning

The font size of the following Simulink block or signal names are different from input parameters:

- sldemo_absbrake/annotation

Recommended Action

Consider modifying font size of block and signal names as per input parameters.

 Check whether block names appear below blocks (28-Apr-2026 21:52:27)
Incorrect block name position

Warning

The following blocks have names that do not display below the blocks:

- sldemo_absbrake/Vehicle Dynamics/Unit Delay

Recommended Action

Change the location such that the block name is below the block.

 Check the display attributes of block names (28-Apr-2026 21:52:27)
Check for missing block names

Identify block names that are hidden but should be displayed to show a descriptive name.

Warning

The following blocks have descriptive names, however, the names are hidden:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table
- sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table1
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Bus Creator
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Step
- sldemo_absbrake/Vehicle Dynamics/Vehicle/-1//m
- sldemo_absbrake/Visualization/Gain
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Rate Transition
- sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table2
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Stop Simulation2
- sldemo_absbrake/Zero-Order Hold1
- sldemo_absbrake/S-R Flip-Flop
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Rr
- sldemo_absbrake/Zero-Order Hold

^ Less

Recommended Action

Modify the block to show the block name by selecting (**Format > Auto Name > Name On**).

 Check for display of block parameter (28-Apr-2026 21:52:27)

Check for display of block parameter

Identify block parameter that are not displayed in the model.

Warning

Following blocks do not display block parameter:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/-1//m
- sldemo_absbrake/ctrl
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Weight
- sldemo_absbrake/Target Slip
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle speed (angular)
- sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table2
- sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table1
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Rate Transition
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle Speed
- sldemo_absbrake/Subtract
- sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table
- sldemo_absbrake/Switch
- sldemo_absbrake/Dry Mode
- sldemo_absbrake/Always Latch
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Rr
- sldemo_absbrake/Visualization/Gain
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Stopping distance
- sldemo_absbrake/Vehicle Dynamics/Unit Delay
- sldemo_absbrake/Ice Mode

^ Less

Recommended Action

For the above blocks, display the block parameter in the Block Annotation pane of the Block Properties dialog box.

 Check Model Description (28-Apr-2026 21:52:27)
Identify layers in the model having inconsistent description format.

Warning

Following layers do not have consistent model description format:

- sldemo_absbrake

Recommended Action

Consider having a consistent format for the model description

Example: If description tags are 'Input:', 'Description:', and 'Output:' then format should be as following:

Input: add input information here

Description: add model description here

Output: add output information here

layers in the model having inconsistent description format. Identify

Warning

Following layers do not have model descriptions:

- sldemo_absbrake/Vehicle Dynamics/Vehicle
- sldemo_absbrake/Visualization
- sldemo_absbrake/Vehicle Dynamics/Relative Slip
- sldemo_absbrake/Vehicle Dynamics

Recommended Action

Consider adding model description for all the layers.

 Check if blocks are shaded in the model (28-Apr-2026 21:52:27)

Check if blocks are shaded in the model

Check for shaded blocks in the model.

Warning

The following blocks in the model are shaded

- sldemo_absbrake/More Info

Recommended Action

Turn off the 'DropShadow' property of shaded blocks for better readability.

 Check for unconnected signal lines and blocks (28-Apr-2026 21:52:27)

Check for unconnected subsystems and basic blocks

Unconnected basic blocks and subsystems.

Warning

The following blocks in the model are not connected:

- sldemo_absbrake/S-R Flip-Flop

Recommended Action

Connect the blocks to the correct source or destination block.

If the destination block is not known, use a Terminator or Ground block to terminate the line.

✓ Check signal line connections (28-Apr-2026 21:52:27)
Identify intersections and overlaps of signals in a model.

Passed

No signal intersections or overlaps found.

⚠ Check signal flow in model (28-Apr-2026 21:52:27)
Check placement of sequential blocks

Warning

The placement of blocks in the following subsystems can be improved:

- sldemo_absbrake/Vehicle Dynamics/Vehicle
- sldemo_absbrake

Recommended Action

Ensure that the signal flow in the mentioned subsystems is from left to right.

- All sequential blocks, except the blocks on feedback path, must be placed from left to right.
- All blocks, except the blocks on feedback path, should be oriented to the right.

⚠ Check usage of tunable parameters in blocks (28-Apr-2026 21:52:27)
Identify tunable parameters used to specify expressions, data type conversions, or indexing operations.

Warning

The following blocks have tunable parameters that specify expressions, data type conversions, or indexing operations:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle speed (angular)
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Weight
- sldemo_absbrake/Vehicle Dynamics/Vehicle/-1//m

Recommended Action

Modify the use of the tunable parameters in one of the following methods:

- Perform the calculations using Simulink basic blocks or precompute the values.
- Use the Selector block to extract array entries.
- Use the Data Type Conversion block to change data types.

 [Check connections between structural subsystems \(28-Apr-2026 21:52:27\)](#)
Identify connections between structural subsystems.

Passed

All connections to structural subsystems adhere to the guideline.

 [Check for consistency in model element names \(28-Apr-2026 21:52:27\)](#)
Check for consistency in model element names

Naming mismatch with signal name 'Wv'

Warning

Naming mismatch with signal name 'Wv'

- sldemo_absbrake/Vehicle Dynamics/Ww
- sldemo_absbrake/Visualization/Ww

Recommended Action

Consider renaming the deviating model elements to match the signal name or to be consistent with Inport/Outport blocks.

Check

for consistency in model element names

Naming mismatch with block 'Ww'

Warning

- sldemo_absbrake/Visualization
- sldemo_absbrake/Wv

Recommended Action

Consider renaming the deviating model elements to match the signal name or to be consistent with Inport/Outport blocks.

Check

for consistency in model element names

Naming mismatch with signal name 'slp'

Warning

Naming mismatch with signal name 'slp'

- sldemo_absbrake/Vehicle Dynamics/Vehicle/slp_in
- sldemo_absbrake/Vehicle Dynamics/slp_in

Recommended Action

Consider renaming the deviating model elements to match the signal name or to be consistent with Inport/Outport blocks.

Check

for consistency in model element names

Naming mismatch with block 'slp_in'

Warning

- sldemo_absbrake/Vehicle Dynamics/Vehicle/slp
- sldemo_absbrake/slp

Recommended Action

Consider renaming the deviating model elements to match the signal name or to be consistent with Inport/Outport blocks.

✓ Check trigger signal names (28-Apr-2026 21:52:27)

Identify trigger blocks where the origin of the trigger signal and the destination have dissimilar names.

Passed

No violation of the guideline for use of trigger signal names.

⚠ Check for mixing basic blocks and subsystems (28-Apr-2026 21:52:27)

Identify levels in the model that include basic blocks and subsystems. Each level of a model must be designed with blocks of the same level (for example, only subsystems or only basic blocks).

Warning

The following level(s) in the model include basic blocks and subsystems:

System	Block path
sldemo_absbrake	sldemo_absbrake/Always Latch
sldemo_absbrake	sldemo_absbrake/Compare To Constant
sldemo_absbrake	sldemo_absbrake/Dry Mode
sldemo_absbrake	sldemo_absbrake/Ice Mode
sldemo_absbrake	sldemo_absbrake/S-R Flip-Flop
sldemo_absbrake	sldemo_absbrake/Subtract
sldemo_absbrake	sldemo_absbrake/Switch
sldemo_absbrake	sldemo_absbrake/Target Slip
sldemo_absbrake	sldemo_absbrake/Zero-Order Hold
sldemo_absbrake	sldemo_absbrake/Zero-Order Hold1
sldemo_absbrake	sldemo_absbrake/ctrl

^ Less

Recommended Action

If possible, replace blocks at the identified level of the model hierarchy with basic blocks. Move nonvirtual blocks into the identified subsystem.

 Check for avoiding algebraic loops between subsystems (28-Apr-2026 21:52:27)
jc_0653: Delay block layout in feedback loops

Identify delay blocks usage in feedback loops.

Warning

Following delay blocks in feedback loops are placed inside subsystem.

- sldemo_absbrake/Vehicle Dynamics/Unit Delay

Recommended Action

Delay blocks in feedback loops must be placed outside of their parent subsystem.

 Check for prohibited sink blocks (28-Apr-2026 21:52:27)
Prohibited sink blocks

Warning

The following discrete controllers contain sink blocks:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Stop Simulation2

Recommended Action

Remove the prohibited blocks from the subsystem.

 Signal  0  0  6  0  0  0  0

 Check usage of vector and bus signals (28-Apr-2026 21:52:27)
Check bus signals treated as vectors

Warning

The following configuration parameters are set inappropriately:

Parameter	Current Value	Recommended Values
Bus signal treated as vector (StrictBusMsg)	none	error

Recommended Action

Consider setting the configuration parameters to the recommended value.

usage of bus signals

Check

Warning

The following blocks are not supported for use with bus signals:

- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Relative Slip

Recommended Action

Consider remodeling to use buses only with bus-supported blocks.

 Check definition of signal labels (28-Apr-2026 21:52:27)
Check source block labels

Identify source blocks that require labeled signals. A subset of source blocks require labeled signals.

Warning

The following signals have no label:

- sldemo_absbrake
- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake

Recommended Action

Add a new or propagated label to the signal line.

Check

destination block labels

Identify destination blocks that require labeled signals. A subset of destination blocks require labeled signals.

Warning

The following signals have no label:

- sldemo_absbrake

- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake

Recommended Action

Add a new or propagated label to the signal line.

 [Check Signal name propagation \(28-Apr-2026 21:52:27\)](#)

Check Signal name propagation

Signal Name propagation for subsystems.

Warning

The signals from the following subsystems do not have signal propagation turned on:

- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake/Vs
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Ww
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Sd
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake/slp
- sldemo_absbrake
- sldemo_absbrake/Wv
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Vs
- sldemo_absbrake/Vehicle Dynamics/slp
- sldemo_absbrake/Sd
- sldemo_absbrake/Vehicle Dynamics/Vehicle/slp

^ Less

Recommended Action

Enable signal propagation by selecting 'Show propagated signal' parameter for signals.

 [Check position of signal labels \(28-Apr-2026 21:52:27\)](#)

Check location of signal labels

Identify inappropriately placed signal labels.

Warning

The following signals do not have labels located at the origin of the signal line:

- sldemo_absbrake/Sd
- sldemo_absbrake/Vehicle Dynamics/Vehicle/Ff
- sldemo_absbrake/Visualization
- sldemo_absbrake/Vehicle Dynamics/Wa
- sldemo_absbrake/Visualization
- sldemo_absbrake/Visualization
- sldemo_absbrake/Visualization
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/yout
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Ww
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Vs
- sldemo_absbrake/Vehicle Dynamics/Vehicle/vehicle _deceleration
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Sd
- sldemo_absbrake/Wv
- sldemo_absbrake/Vs

^ Less

Recommended Action

Consider placing the labels at the origin of the signal line.

 [Check signal line labels \(28-Apr-2026 21:52:27\)](#)

Check source block labels

Identify source blocks that require labeled signals. A subset of source blocks require labeled signals.

Warning

The following signals have no label:

- sldemo_absbrake
- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake

Recommended Action

Add a new or propagated label to the signal line.

destination block labels

Check

Identify destination blocks that require labeled signals. A subset of destination blocks require labeled signals.

Warning

The following signals have no label:

- sldemo_absbrake
- sldemo_absbrake/Vehicle Dynamics
- sldemo_absbrake
- sldemo_absbrake

Recommended Action

Add a new or propagated label to the signal line.

 Check for propagated signal labels (28-Apr-2026 21:52:27)
Identify propagated labels on signal lines.

Warning

The following signal labels are not propagated. Propagate signals coming from Subsystem blocks.

- sldemo_absbrake/Sd
- sldemo_absbrake/Vehicle Dynamics/slp
- sldemo_absbrake/Wv
- sldemo_absbrake/Vs
- sldemo_absbrake/slp

Recommended Action

Add labels to the output signals.

propagated labels on signal lines.

Identify

Warning

The following subsystem-level Inport block signals should propagate signal labels from the parent system:

- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Sd
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Ww
- sldemo_absbrake/Vehicle Dynamics/Vehicle/slp
- sldemo_absbrake/Vehicle Dynamics/Relative Slip/Vs

Recommended Action

Add labels to the input signals.

 Block Consistency  0  0  3  0  3  0  0

 Check Indexing Mode (28-Apr-2026 21:52:27)

Check for Zero-Based Indexing Mode.

Warning

The following blocks or charts don't have Zero-Based Indexing mode:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Multiport Switch

Recommended Action

Consider remodeling by using Zero-Based Indexing.

✓ Check block orientation (28-Apr-2026 21:52:27)
Passed

⚠ Check if tunable block parameters are defined as named constants (28-Apr-2026 21:52:27)
Check if tunable block parameters are defined as named constants

Warning

The following blocks contain tunable parameters that are not defined as named constants.

Block	Violating Tunable Parameters
sldemo_absbrake/Vehicle Dynamics/Vehicle/Stopping distance	(UpperSaturationLimit = 1000)
sldemo_absbrake/Ice Mode	(Value = 0.15)
sldemo_absbrake/Visualization/Gain	(Gain = 100)
sldemo_absbrake/Vehicle Dynamics/Vehicle/Vehicle Speed	(gainval = 1.0), (UpperSaturationLimit = 1000)
sldemo_absbrake/Dry Mode	(Value = 0.8)

--	--

Recommended Action

Consider changing tunable block parameter literal values to named constants.

-
-  Check for sample time setting (28-Apr-2026 21:52:27)
Check if sample time property of a block is set to -1 (inherited).

Warning

The following blocks do not have sample time set to -1 (inherited):

- sldemo_absbrake/Zero-Order Hold1
- sldemo_absbrake/Zero-Order Hold

Recommended Action

Consider changing the sample time to -1 (inherited).

-
-  Check usage of fixed-point data type with non-zero bias (28-Apr-2026 21:52:27)
Passed

-
-  Check type setting by data objects (28-Apr-2026 21:52:27)
jc_0644: Type setting

Identify blocks that violate signal data type setting if signal objects are used.

Passed

No blocks found that violate signal data type setting if signal objects are used.

- ✓ Check position of conditional blocks and iterator blocks (28-Apr-2026 21:52:27)
Identify conditional and iterative blocks that are positioned inconsistently in the model.

Passed

The conditional and iterative blocks are correctly placed in the model.

- ✓ Check undefined initial output for conditional subsystems (28-Apr-2026 21:52:27)

Passed

- ✓ Check usage of Merge block (28-Apr-2026 21:52:27)

jc_0659: Usage restrictions of signal lines input to Merge blocks

There must not be any block between a Conditional Subsystem block and a Merge block.

Passed

No Merge block found.

- ✓ Check logical expressions in If blocks (28-Apr-2026 21:52:27)

Checks If blocks for complex usage of primary expressions within a logical expression

Passed

Logical expressions inside If blocks are simple

- ✓ Check default/else case in Switch Case blocks and If blocks (28-Apr-2026 21:52:27)

Check if default/else case in Switch Case blocks and If blocks are set to 'on'

Passed

Conditional Control blocks are valid.

- ✓ Check fundamental logical and numerical operations (28-Apr-2026 21:52:27)

Passed

- ✔ Check usage of Sum blocks (28-Apr-2026 21:52:27)
Identify Sum block usage that can affect readability.

Passed

No violations of the guideline found with the usage of the Sum block.

-
- ✔ Check operator order of Product blocks (28-Apr-2026 21:52:27)
Passed

-
- ✔ Check signs of input signals in product blocks (28-Apr-2026 21:52:27)
jc_0611: Input sign for multiplication and division blocks

Identify blocks that perform division whose inputs have different sign bit.

Passed

No product block with division of different sign bits found.

-
- ⚠ Check for parentheses in Fcn block expressions (28-Apr-2026 21:52:27)
Identify order of parentheses in Fcn block expressions.

Warning

The following Fcn blocks do not use parentheses to mark operator precedence:

Block Path	Expression
sldemo_absbrake/Vehicle Dynamics/Relative Slip/Relative Slip	$1.0 - u(1)/(u(2) + (u(2)==0)*eps)$

Recommended Action

Consider resolving the operator precedence in Fcn block expressions by adding parentheses.

✔ Check icon shape of Logical Operator blocks (28-Apr-2026 21:52:27)

Passed

✔ Check usage of Relational Operator blocks (28-Apr-2026 21:52:27)

Identify Relational Operator blocks that connect to constants with the first (upper) input value.

Passed

This model does not contain Relational Operator blocks.

✔ Check comparison of floating point types in Simulink (28-Apr-2026 21:52:27)

jc_0800: Comparing floating-point types in Simulink

Equivalence comparison should not be used for floating point numbers.

Passed

No Equivalence comparison done on floating point numbers.

⚠ Check usage of Lookup Tables (28-Apr-2026 21:52:27)

Check Lookup method settings of n-D Lookup table blocks

Warning

The following n-D Lookup table blocks violate recommended Lookup method settings:

Block	Parameter	Current Value	Recommended Values
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table1	UseLastTableValue	off	on
sldemo_absbrake/Target Slip	UseLastTableValue	off	on

sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table	ExtrapMethod	Linear	Clip
sldemo_absbrake/Target Slip	ExtrapMethod	Linear	Clip
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table	UseLastTableValue	off	on
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table1	ExtrapMethod	Linear	Clip
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table2	UseLastTableValue	off	on
sldemo_absbrake/Vehicle Dynamics/Vehicle/1-D Lookup Table2	ExtrapMethod	Linear	Clip

Recommended Action

Consider changing the above mentioned block parameters with the recommended values.

 Check usage of Memory and Unit Delay blocks (28-Apr-2026 21:52:27)

Identify Memory blocks not using a continuous sample time

Passed

No Memory blocks found with inappropriate sample time

Identify Unit Delay blocks with non-discrete sample time

Warning

The following Unit Delay blocks have non-discrete sample time:

- sldemo_absbrake/Vehicle Dynamics/Unit Delay

Recommended Action

Consider using Memory block instead.

✔ Check for cascaded Unit Delay blocks (28-Apr-2026 21:52:27)

Identify cascaded and tapped pattern of Unit Delay blocks.

Passed

No cascaded Unit Delay blocks found that can be changed to Tapped Delay/Delay block.

✔ Check usage of Discrete-Time Integrator block (28-Apr-2026 21:52:27)

Check usage of recommended settings for Discrete-Time Integrator blocks to prevent unexpected results.

Passed

All Discrete-Time Integrator blocks have recommended settings.

✔ Check usage of the Saturation blocks (28-Apr-2026 21:52:27)

jc_0628: Usage of Saturation blocks

Identify the Saturation and Saturation Dynamic blocks that perform type casting.

Passed

No Saturation and/or Saturation Dynamic blocks perform type casting

⚠ Check output data type of operation blocks (28-Apr-2026 21:52:27)

Check output data type of operation blocks

Identify operation blocks that directly specify the 'Output data type' parameter when changing the data type of the block output signal.

Warning

The following operation blocks directly specify the 'Output data type' parameter:

- sldemo_absbrake/Compare To Constant

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Step

Recommended Action

Instead of directly specifying the 'Output data type' parameter on operation blocks, use Data Type Conversion blocks.

 Check for division by zero in Simulink (28-Apr-2026 21:52:27)

Error occurred during Simulink Design Verifier analysis. Model 'sldemo_absbrake' has referenced models with inconsistent Hardware Implementation settings for 'ProdEndianness' parameter. Simulink Design Verifier does not support this. Simulink Design Verifier failed to initialize: 'sldemo_absbrake' is incompatible for design error detection with Simulink Design Verifier.

 Other blocks  0  0  2  0  8  0  0

 Check position of Inport and Outport blocks (28-Apr-2026 21:52:27)
Identify inappropriately placed Inport and Outport blocks.

Passed

No Inport or Outport blocks found which are inappropriately placed.

 Check display for port blocks (28-Apr-2026 21:52:27)
Passed

 Check scope of From and Goto blocks (28-Apr-2026 21:52:27)
Passed

 Check for usage of Data Store Memory blocks (28-Apr-2026 21:52:27)
Identify the usage of Data Store Memory blocks.

Passed

Usage of Data Store Memory blocks is correct.

 Check usage of Switch blocks (28-Apr-2026 21:52:27)

Check usage of Switch blocks

Identify Switch blocks with the parameter **Criteria for passing first input** not set to $u2 \sim 0$.

Warning

The following Switch blocks do not have the parameter **Criteria for passing first input** set to $u2 \sim 0$

- sldemo_absbrake/Switch

Recommended Action

Set the block parameter **Criteria for passing first input** to $u2 \sim 0$. This might require reworking the logic associated with the Switch block.

 Check input and output data type for Switch blocks (28-Apr-2026 21:52:27)

Passed

 Check settings for data ports in Multiport Switch blocks (28-Apr-2026 21:52:27)

Check input type to Multiport Switch block

Warning

The following Multiport Switch blocks have incorrect input type:

- sldemo_absbrake/Vehicle Dynamics/Vehicle/Multiport Switch

Recommended Action

Consider changing the input type to unsigned integer.

- ✓ Check for missing ports in Variant Subsystems (28-Apr-2026 21:52:27)
Check for number of inputs/outputs to a Variant Subsystem.

Passed

No Variant Subsystems found having different number of inputs/outputs in the Variant Subsystem choices.

-
- ✓ Check use of default variants (28-Apr-2026 21:52:27)
na_0036: Default variant

Identify variant subsystems that do not use default variants.

Passed

All variant subsystems in the model use default variants

-
- ✓ Check use of single variable variant conditionals (28-Apr-2026 21:52:27)
Identify variant subsystems which use multi-variable compound conditions.

Passed

No variant subsystems with multiple variable compound conditions found

📁 Stateflow 0 0 2 0 47 0 0

📁 Block/Data/Events 0 0 0 0 6 0 0

-
- ✓ Check for names of Stateflow ports and associated signals (28-Apr-2026 21:52:27)

Identify mismatches between names of Stateflow ports and the associated signals.

Passed

No Stateflow charts were found.

-
- ✓ Check execution timing for default transition path (28-Apr-2026 21:52:27)
'Execute (enter) Chart At Initialization' should be set to OFF.

Passed

All Stateflow Charts pass the check.

✔ Check definition of Stateflow data (28-Apr-2026 21:52:27)

Check definition of Stateflow data

Passed

All Stateflow data at machine level has been defined as per guideline.

✔ Check usable number for first index (28-Apr-2026 21:52:27)

Identify usage of first index of Stateflow data.

Passed

All Stateflow data first index values are uniform.

✔ Check scope of data in parallel states (28-Apr-2026 21:52:27)

jc_0722: Local data definition in parallel states

The scope of local variables should be restricted to one parallel state unless it is being used by other parallel states.

Passed

No Stateflow States were found.

✔ Check definition of Stateflow events (28-Apr-2026 21:52:27)

Stateflow events should be defined at the smallest possible scope of usage.

Passed

All Stateflow events are defined at their smallest scope.

 Diagram 0 0 2 0 12 0 0

✔ Check for unconnected objects in Stateflow Charts (28-Apr-2026 21:52:27)

Identify dangling transitions and unconnected Stateflow States and Junctions in Stateflow Charts.

Passed

No unconnected transitions, states or junctions found in Stateflow Charts.

✔ Check for exclusive states in state machines (28-Apr-2026 21:52:27)
Passed

✔ Check usage of parallel states (28-Apr-2026 21:52:27)
Passed

✔ Check Stateflow transition appearance (28-Apr-2026 21:52:27)
Identify Stateflow transitions visually overlapping other Stateflow objects.

Passed
No transition violates the guidelines for Stateflow transition appearance.

⚠ Check default transition placement in Stateflow charts (28-Apr-2026 21:52:27)
Check for unconditional default transitions.

Warning

The following default transitions do not have a single non-guard path to any state at the level in the chart:

Parameter	Current Value	Recommended Values
No unconditional default transitions (SFNoUnconditionalDefaultTransitionDiag)	warning	error

Recommended Action

Set 'No unconditional default transitions' parameter to recommended value 'error'.

✔ Check usage of transitions to external states (28-Apr-2026 21:52:27)
Passed

⚠ Check for unexpected backtracking in state transitions (28-Apr-2026 21:52:27)
Identify configuration parameter settings which identify unexpected backtracking in state transitions.

Warning

The model configuration parameters are not set to the recommended values specified in the data file.

Status	Parameter	Current Value	Recommended Values
Warning	Unexpected backtracking (SFUnexpectedBacktrackingDiag)	warning	error

Recommended Action

Follow the links in the result table to modify the model configuration parameters.

✔ Check usage of internal transition (28-Apr-2026 21:52:27)
Passed

✔ Check usage of internal transitions in Stateflow states (28-Apr-2026 21:52:27)
Identify Stateflow states using multiple internal transitions.

Passed

No Stateflow states found with multiple internal transitions

✔ Check prohibited combination of state action and flow chart (28-Apr-2026 21:52:27)
Passed

✔ Check transitions in Stateflow flow charts (28-Apr-2026 21:52:27)
Passed

✔ Check usage of unconditional transitions in flow charts (28-Apr-2026 21:52:27)
Identify unconditional transitions in flow charts.

Passed
All unconditional transitions adhere to the guideline.

✔ Check terminal junctions in Stateflow (28-Apr-2026 21:52:27)
Identify usage of terminal junctions in flow charts.

Passed
Multiple terminal junctions were not found.

✔ Check usage of Stateflow comments (28-Apr-2026 21:52:27)
Identify comments that are nested or contain newline(s) in the middle in Stateflow for action language 'C'.

Passed
No comments found that are either nested or contain newline(s) in the middle.

📁 Condition Transition/Action 0 0 0 0 17 0 0

✔ Check Stateflow chart action language (28-Apr-2026 21:52:27)
Passed

✔ Check usage of numeric literals in Stateflow (28-Apr-2026 21:52:27)
Identify use of numeric literals in Stateflow states and transitions.

Passed
No numeric literals found in Stateflow charts.

- ✔ Check for pointers in Stateflow charts (28-Apr-2026 21:52:27)
jm_0011: Pointers in Stateflow

Identify pointer operations on custom code variables.

Passed

No pointer operations were found.

- ✔ Check usage of events in Stateflow charts (28-Apr-2026 21:52:27)
Identify undirected event broadcasts in Stateflow.

Passed

No instances of undirected event broadcast were found.

- ✔ Check order of state action types (28-Apr-2026 21:52:27)
Identify out of order state action types in Stateflow states.

Passed

No Stateflow states found with out of order state action types

- ✔ Check repetition of Action types (28-Apr-2026 21:52:27)
jc_0734: Number of state action types
Identifies repeated action types in a Stateflow State.

Passed

No Stateflow States were found.

- ✔ Check if state action type 'exit' is used in the model (28-Apr-2026 21:52:27)
Check if state action type 'exit' is used in the model.

Passed

State action type 'exit' is not used in the model.

- ✔ Check updates to variables used in state transition conditions (28-Apr-2026 21:52:27)
jc_0741: Timing to update data used in state chart transition conditions

Variables used in state transition conditions must not perform an update by "during" state action type.

Passed

No Stateflow states found that violate the guidelines for updating the variables used in state transition conditions.

- ✔ Check usage of transition conditions in Stateflow transitions (28-Apr-2026 21:52:27)
Identify unconditional Stateflow transitions with higher priority than conditional transitions.

Passed

No unconditional Stateflow transitions found with higher priority than conditional transitions

- ✔ Check condition actions and transition actions in Stateflow (28-Apr-2026 21:52:27)
Identify usage of transition actions in Stateflow.

Passed

No Stateflow charts have transition actions.

- ✔ Check for MATLAB expressions in Stateflow blocks (28-Apr-2026 21:52:27)

Passed

- ✔ Check usage of floating-point expressions in Stateflow charts (28-Apr-2026 21:52:27)
Identify equality or inequality operations (`==`, `~=`, `!=`, `<>`) in expressions involving floating-point variables or constants.

Passed

No equality or inequality operations in expressions where at least one side of the expression is a floating-point variable or constant were found.

- ✔ Check Stateflow operators (28-Apr-2026 21:52:27)
Identify the usage of operators in Stateflow.

Passed

No Stateflow blocks found with incorrect operator usage.

- ✔ Check prohibited comparison operation of logical type signals (28-Apr-2026 21:52:27)
Identify boolean variables in Stateflow charts using comparison operations.

Passed

No boolean variables use comparison operations of logical type in the model.

- ✓ Check usage of unary minus operations in Stateflow charts (28-Apr-2026 21:52:27)
Identify unary minus operations applied to unsigned integers in Stateflow objects.

Passed

No unary minus operations applied to unsigned integers in Stateflow objects were found.

- ✓ Check for implicit type casting in Stateflow (28-Apr-2026 21:52:27)
Identify implicit type casting in Stateflow.

Passed

No instances of implicit type casting found.

- ✓ Check usage of graphical functions in Stateflow (28-Apr-2026 21:52:27)
Check for calls between graphical functions.

Passed

No calls between graphical functions were found.

 Label Description  0  0  0  9  0  0

- ✓ Check uniqueness of Stateflow State, Data and Event names (28-Apr-2026 21:52:27)
jc_0732: Distinction between state names, data names, and event names
Identify Stateflow State, Stateflow Data and Stateflow Event that have identical names in a given chart.

Passed

No Stateflow charts were found.

- ✓ Check uniqueness of State names (28-Apr-2026 21:52:27)
jc_0730: Unique state name in Stateflow blocks
Identifies identical State names within a Stateflow Chart.

Passed

No Stateflow charts were found.

-
- ✓ Check usage of State names (28-Apr-2026 21:52:27)
jc_0731: State name format

Identify state names with '/' at its end.

Passed

No Stateflow states were found.

-
- ✓ Check entry formatting in State blocks in Stateflow charts (28-Apr-2026 21:52:27)
Passed

-
- ✓ Check indentation of code in Stateflow states (28-Apr-2026 21:52:27)
Identify non-uniform indentation in Stateflow blocks.

Passed

All Stateflow blocks have uniform indentation.

-
- ✓ Check position of label string in Stateflow transition (28-Apr-2026 21:52:27)
Identify placement of label string in Stateflow transition.

Passed

All Stateflow transitions are placed uniformly.

-
- ✓ Check position of comments in transition labels (28-Apr-2026 21:52:27)
Identify comments in transition labels that are not positioned uniformly.

Passed

Comments in transition labels are positioned uniformly.

-
- ✓ Check usage of parentheses in Stateflow transitions (28-Apr-2026 21:52:27)
jc_0752: Condition action in transition label

Start new line before and after parentheses for condition actions in Stateflow transitions.

Passed

No Stateflow Transitions found that violate the requirement for new line for condition actions.

-
- ✓ Check for comments in unconditional transitions (28-Apr-2026 21:52:27)
Identify comments in unconditional transitions without action statements.

Passed

All unconditional transitions without action statements have comments.

📁 Miscellaneous 0 0 0 0 3 0 0

-
- ✓ Check return value assignments in Stateflow graphical functions (28-Apr-2026 21:52:27)
Identify graphical functions with multiple assignments of return values in Stateflow charts.

Passed

No Stateflow charts were found.

-
- ✓ Check usage of Simulink functions in Stateflow (28-Apr-2026 21:52:27)
Usage of Simulink Functions in Stateflow.

Passed

All Simulink Functions in Stateflow are defined according to the guideline.

-
- ✓ Check use of Simulink in Stateflow charts (28-Apr-2026 21:52:27)
na_0039: Limitation on Simulink functions in Chart blocks

Check use of Stateflow charts nested inside Simulink functions used in Stateflow.

Passed

No Stateflow charts found nested inside Simulink functions used in Stateflow.

📁 MATLAB 0 0 0 0 20 0 0

📁 Appearance 0 0 0 0 3 0 0

-
- ✓ Check nested conditions in MATLAB Functions (28-Apr-2026 21:52:27)
Identify nested if/else and case statements in MATLAB Functions.

Passed

No MATLAB Function found with deeply nested if/else and case statements.

- ✓ Check for spaces in function or variable names (28-Apr-2026 21:52:27)
Identify single-byte spaces between function or variable names and left parenthesis symbol.

Passed

No spaces between function or variable names and left parenthesis symbol.

- ✓ Check function headers (28-Apr-2026 21:52:27)
Identify functions without a function header.

Passed

All functions have a function header.

 Data and Operations 0 0 0 0 4 0 0

- ✓ Check MATLAB code for global variables (28-Apr-2026 21:52:27)

Check for global variables in MATLAB code

Check for global variables in MATLAB code used in MATLAB Function blocks

Passed

No MATLAB Function blocks found

Check for global variables in MATLAB functions defined in Stateflow charts

Passed

No MATLAB functions defined in Stateflow charts found

Check for global variables in called MATLAB functions

Passed

No external MATLAB functions found

✔ Check usage of enumerated values (28-Apr-2026 21:52:27)

Identify enumeration classes used in the model with no default value specification.

Passed

No enumeration classes found without default value specifications.

✔ Check input and output settings of MATLAB Functions (28-Apr-2026 21:52:27)

Identify MATLAB Functions that have inputs, outputs, or parameters with inherited complexity, data type, or size properties.

Passed

No MATLAB Functions found in the model or subsystem.

✔ Check number of lines of functions (28-Apr-2026 21:52:27)

Identify functions that contain more lines than the threshold value.

Passed

The number of lines of all functions is less than or equal to the threshold value.

 Usage 0 0 0 0 13 0 0

✔ Check lines of code in MATLAB Functions (28-Apr-2026 21:52:27)

Identify MATLAB Functions with high number of effective lines of code.

Passed

No MATLAB Function found with high number of effective lines of code.

✔ Check the number of function calls in MATLAB Function blocks (28-Apr-2026 21:52:27)

Checks whether number of function calls in MATLAB Function blocks is less than 3.

Passed

Number of function calls in MATLAB Function blocks is less than 3.

✔ Check usage of character vector inside MATLAB Function block (28-Apr-2026 21:52:27)

Identify usage of strings in MATLAB Function blocks.

Passed

No character vectors found in MATLAB Function block

✔ Check usage of recommended patterns for Switch/Case statements (28-Apr-2026 21:52:27)

Identify usage of non-constant variables in Switch/Case statements.

Passed

Non-constant variables are not used as Switch/Case arguments

✔ Check for use of C-style comment symbols (28-Apr-2026 21:52:27)

Identify usage of C-style comments in CGT Files and MPT Objects.

Passed

C-style comments are not used in CGT Files and MPT Objects.

✔ Check description of execution statements (28-Apr-2026 21:52:27)

Identify multiple execution statements that are written in a single line and end without a semicolon.

Passed

All execution statements are written in new lines and end with a semicolon.

✔ Check for operator precedence (28-Apr-2026 21:52:27)

Identify expressions that contain operators with different priorities but do not specify clear operator precedence using parentheses.

Passed

All expressions containing operators with different priorities specify clear operator precedence using parentheses.

✔ Check spaces in expressions (28-Apr-2026 21:52:27)

Identify missing spaces before or after binary operators or after commas.
Identify invalid spaces between unary operators and operands.

Passed

All expressions have a space before and after binary operators and after commas, and no space between unary operators and operands.

✔ Check description of conditional expressions (28-Apr-2026 21:52:27)

Identify conditional expressions that do not evaluate to a single logical value or are written in a complex or non-uniform way.

Passed

All conditional expressions evaluate to a single logical value and are not written in a complex or non-uniform way.

✔ Check relational operator usage (28-Apr-2026 21:52:27)

Check for correct placement of relational operator operands.

Passed

All relational operator usage is valid.

✔ Check for utilization of the return value of functions (28-Apr-2026 21:52:27)

Identify function return values that are used before assigning to a variable.

Passed

Return value of all functions used was assigned to a variable before using it.

✔ Check array indices (28-Apr-2026 21:52:27)

Identify array indices calculated by an expression.

Passed

Array indices must not be calculated by an expression.

✔ Check for usage of non-empty statements (28-Apr-2026 21:52:27)

Identify non-empty statements that do not have any effect or do not change the control flow.

Passed

All non-empty statements have a side-effect or change the control flow.

 Modeling Standards for JMAAB v5.1 0 0 0 0 0 0  132

 Naming Conventions 0 0 0 0 0 0  14

 Check file names

Not Run

 Check folder names

Not Run

 Check subsystem names

Not Run

 Check port block names

Not Run

 Check character usage in block names

Not Run

 Check character usage in signal names and bus names

Not Run

 Check character usage in parameter names

Not Run

 Check length of model file name

Not Run

 Check length of folder name at every level of model path
Not Run

 Check length of subsystem names
Not Run

 Check length of Inport and Outport names
Not Run

 Check length of signal and bus names
Not Run

 Check length of parameter names
Not Run

 Check length of block names
Not Run

 *Model Architecture* 0 0 0 0 0 1

 Check for mixing basic blocks and subsystems
Not Run

 *Model Configuration Options* 0 0 0 0 0 2

 Check Implement logic signals as Boolean data (vs. double)
Not Run

 Check diagnostic settings for incorrect calculation results
Not Run

 Check for Simulink diagrams using nonstandard display attributes
Not Run

 Check Model font settings
Not Run

 Check position of Inport and Outport blocks
Not Run

 Check whether block names appear below blocks
Not Run

 Check the display attributes of block names
Not Run

 Check for display of block parameter
Not Run

 Check trigger signal names
Not Run

 Check for unconnected signal lines and blocks
Not Run

 Check usage of Switch blocks
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check Indexing Mode
Not Run

 Check usage of tunable parameters in blocks
Not Run

 Check definition of signal labels
Not Run

 Check Signal name propagation
Not Run

 Check usage of Discrete-Time Integrator block
Not Run

 Check settings for data ports in Multiport Switch blocks
Not Run

 Check usage of fixed-point data type with non-zero bias
Not Run

 Check input and output data type for Switch blocks
Not Run

 Check signs of input signals in product blocks
Not Run

 Check Signed Integer Division Rounding mode
Not Run

 Check type setting by data objects
Not Run

 Check usage of the Saturation blocks
Not Run

 Check usage of Merge block
Not Run

 Check usage of Memory and Unit Delay blocks
Not Run

 Check block orientation
Not Run

 Check if blocks are shaded in the model
Not Run

 Check operator order of Product blocks
Not Run

 Check icon shape of Logical Operator blocks
Not Run

 Check if tunable block parameters are defined as named constants
Not Run

 Check default/else case in Switch Case blocks and If blocks
Not Run

 Check usage of Lookup Tables
Not Run

 Check for parentheses in Fcn block expressions
Not Run

 Check undefined initial output for conditional subsystems
Not Run

 Check for avoiding algebraic loops between subsystems
Not Run

 Check comparison of floating point types in Simulink
Not Run

 Check duplication of Simulink Data names
Not Run

 Check unused data in Simulink Model
Not Run

 Check output data type of operation blocks
Not Run

 Check Model Description
Not Run

 Check for consistency in model element names
Not Run

 Check for sample time setting
Not Run

 Check usage of Sum blocks
Not Run

 Check position of signal labels
Not Run

 Check for missing ports in Variant Subsystems
Not Run

 Check for cascaded Unit Delay blocks
Not Run

 Check for usage of Data Store Memory blocks
Not Run

 Check fundamental logical and numerical operations
Not Run

 Check signal flow in model
Not Run

 Check usage of vector and bus signals
Not Run

 Check connections between structural subsystems
Not Run

 Check position of conditional blocks and iterator blocks
Not Run

 Check signal line connections
Not Run

 Check scope of From and Goto blocks
Not Run

 Check for division by zero in Simulink
Not Run

 Check use of single variable variant conditionals
Not Run

 Check logical expressions in If blocks
Not Run

 Check display for port blocks
Not Run

 Check use of default variants
Not Run

 Stateflow 0 0 0 0 0 0 53

 Check transitions in Stateflow flow charts
Not Run

 Check return value assignments in Stateflow graphical functions
Not Run

 Check entry formatting in State blocks in Stateflow charts
Not Run

 Check default transition placement in Stateflow charts
Not Run

 Check definition of Stateflow data
Not Run

 Check for MATLAB expressions in Stateflow blocks
Not Run

 Check for pointers in Stateflow charts
Not Run

 Check Stateflow operators
Not Run

 Check usage of unary minus operations in Stateflow charts
Not Run

 Check usage of Stateflow comments
Not Run

 Check prohibited comparison operation of logical type signals
Not Run

 Check usage of internal transitions in Stateflow states
Not Run

 Check usage of transition conditions in Stateflow transitions
Not Run

 Check uniqueness of Stateflow State, Data and Event names
Not Run

 Check uniqueness of State names
Not Run

 Check usage of parentheses in Stateflow transitions
Not Run

 Check prohibited combination of state action and flow chart
Not Run

 Check condition actions and transition actions in Stateflow
Not Run

 Check usable number for first index
Not Run

 Check usage of State names
Not Run

 Check execution timing for default transition path
Not Run

 Check repetition of Action types
Not Run

 Check for unused data in Stateflow Charts
Not Run

 Check updates to variables used in state transition conditions
Not Run

 Check usage of internal transition
Not Run

 Check usage of parallel states
Not Run

 Check scope of data in parallel states
Not Run

 Check indentation of code in Stateflow states
Not Run

 Check for usage of text inside states
Not Run

 Check for unexpected backtracking in state transitions
Not Run

 Check for unconnected objects in Stateflow Charts
Not Run

 Check position of label string in Stateflow transition
Not Run

 Check Stateflow chart action language
Not Run

 Check character usage in Stateflow data names
Not Run

 Check length of Stateflow data name
Not Run

 Check usage of transitions to external states
Not Run

 Check order of state action types
Not Run

 Check usage of numeric literals in Stateflow
Not Run

 Check position of comments in transition labels
Not Run

 Check terminal junctions in Stateflow
Not Run

 Check for implicit type casting in Stateflow
Not Run

 Check usage of graphical functions in Stateflow
Not Run

 Check if state action type 'exit' is used in the model
Not Run

 Check for use of C-style comment symbols
Not Run

 Check usage of unconditional transitions in flow charts
Not Run

 Check for comments in unconditional transitions
Not Run

 Check definition of Stateflow events
Not Run

 Check Stateflow transition appearance
Not Run

 Check usage of events in Stateflow charts
Not Run

 Check usage of Simulink functions in Stateflow
Not Run

 Check for exclusive states in state machines
Not Run

 Check usage of floating-point expressions in Stateflow charts
Not Run

 Check use of Simulink in Stateflow charts
Not Run

 MATLAB Functions 0 0 0 0 0 0 4

 Check input and output settings of MATLAB Functions
Not Run

 Check MATLAB code for global variables
Not Run

 Check usage of character vector inside MATLAB Function block
Not Run

 Check usage of enumerated values
Not Run

 Modeling Standards for JMAAB v6 0 0 0 0 0 0  152

 *Naming Conventions* 0 0 0 0 0 0  18

 Check file names
Not Run

 Check folder names
Not Run

 Check subsystem names
Not Run

 Check port block names
Not Run

 Check character usage in block names
Not Run

 Check character usage in signal names and bus names
Not Run

 Check character usage in parameter names
Not Run

 Check length of model file name
Not Run

 Check length of folder name at every level of model path
Not Run

 Check length of subsystem names
Not Run

 Check length of Inport and Outport names
Not Run

 Check length of signal and bus names
Not Run

 Check length of parameter names
Not Run

 Check length of block names
Not Run

 Check data names in MATLAB Functions
Not Run

 Check the length of data names in MATLAB Functions
Not Run

 Check Bus and Enumeration data type names
Not Run

 Check length of Bus and Enumeration data type names
Not Run

 Model Architecture 0 0 0 0 0 0 1

 Check for mixing basic blocks and subsystems
Not Run

 *Model Configuration Options* 0 0 0 0 0 0 1

 Check Implement logic signals as Boolean data (vs. double)
Not Run

 *Simulink* 0 0 0 0 0 0 59

 Check for Simulink diagrams using nonstandard display attributes
Not Run

 Check Model font settings
Not Run

 Check position of Inport and Outport blocks
Not Run

 Check whether block names appear below blocks
Not Run

 Check the display attributes of block names
Not Run

 Check for display of block parameter
Not Run

 Check trigger signal names
Not Run

 Check for unconnected signal lines and blocks
Not Run

 Check usage of Switch blocks
Not Run

 Check usage of Relational Operator blocks
Not Run

 Check Indexing Mode
Not Run

 Check usage of tunable parameters in blocks
Not Run

 Check definition of signal labels
Not Run

 Check Signal name propagation
Not Run

 Check usage of Discrete-Time Integrator block
Not Run

 Check settings for data ports in Multiport Switch blocks
Not Run

 Check usage of fixed-point data type with non-zero bias
Not Run

 Check input and output data type for Switch blocks
Not Run

 Check signs of input signals in product blocks
Not Run

 Check Signed Integer Division Rounding mode
Not Run

 Check type setting by data objects
Not Run

 Check usage of the Saturation blocks
Not Run

 Check usage of Merge block
Not Run

 Check usage of Memory and Unit Delay blocks
Not Run

 Check block orientation
Not Run

 Check if blocks are shaded in the model
Not Run

 Check operator order of Product blocks
Not Run

 Check icon shape of Logical Operator blocks
Not Run

 Check if tunable block parameters are defined as named constants
Not Run

 Check default/else case in Switch Case blocks and If blocks
Not Run

 Check usage of Lookup Tables
Not Run

 Check for parentheses in Fcn block expressions
Not Run

 Check undefined initial output for conditional subsystems
Not Run

 Check for avoiding algebraic loops between subsystems
Not Run

 Check comparison of floating point types in Simulink
Not Run

 Check duplication of Simulink Data names
Not Run

 Check unused data in Simulink Model
Not Run

 Check output data type of operation blocks
Not Run

 Check Model Description
Not Run

 Check for consistency in model element names
Not Run

 Check for sample time setting
Not Run

 Check usage of Sum blocks
Not Run

 Check position of signal labels
Not Run

 Check for missing ports in Variant Subsystems
Not Run

 Check for cascaded Unit Delay blocks
Not Run

 Check for usage of Data Store Memory blocks
Not Run

 Check fundamental logical and numerical operations
Not Run

 Check signal flow in model
Not Run

 Check usage of vector and bus signals
Not Run

 Check connections between structural subsystems
Not Run

 Check position of conditional blocks and iterator blocks
Not Run

 Check signal line connections
Not Run

 Check scope of From and Goto blocks
Not Run

 Check for division by zero in Simulink
Not Run

 Check use of single variable variant conditionals
Not Run

 Check logical expressions in If blocks
Not Run

 Check display for port blocks
Not Run

 Check use of default variants
Not Run

 Check for overlapping or intersecting blocks and signal lines
Not Run

 Stateflow 0 0 0 0 0 55

 Check transitions in Stateflow flow charts
Not Run

 Check return value assignments in Stateflow graphical functions
Not Run

 Check entry formatting in State blocks in Stateflow charts
Not Run

 Check default transition placement in Stateflow charts
Not Run

 Check definition of Stateflow data
Not Run

 Check for MATLAB expressions in Stateflow blocks
Not Run

 Check for pointers in Stateflow charts
Not Run

 Check Stateflow operators
Not Run

 Check usage of unary minus operations in Stateflow charts
Not Run

 Check usage of Stateflow comments
Not Run

 Check prohibited comparison operation of logical type signals
Not Run

 Check usage of internal transitions in Stateflow states
Not Run

 Check usage of transition conditions in Stateflow transitions
Not Run

 Check uniqueness of Stateflow State, Data and Event names
Not Run

 Check uniqueness of State names
Not Run

 Check usage of parentheses in Stateflow transitions
Not Run

 Check prohibited combination of state action and flow chart
Not Run

 Check condition actions and transition actions in Stateflow
Not Run

 Check usable number for first index
Not Run

 Check usage of State names
Not Run

 Check execution timing for default transition path
Not Run

 Check repetition of Action types
Not Run

 Check for unused data in Stateflow Charts
Not Run

 Check updates to variables used in state transition conditions
Not Run

 Check usage of internal transition
Not Run

 Check usage of parallel states
Not Run

 Check scope of data in parallel states
Not Run

 Check indentation of code in Stateflow states
Not Run

 Check for unexpected backtracking in state transitions
Not Run

 Check for unconnected objects in Stateflow Charts
Not Run

 Check position of label string in Stateflow transition
Not Run

 Check Stateflow chart action language
Not Run

 Check character usage in Stateflow data names
Not Run

 Check length of Stateflow data name
Not Run

 Check usage of transitions to external states
Not Run

 Check order of state action types
Not Run

 Check usage of numeric literals in Stateflow
Not Run

 Check position of comments in transition labels
Not Run

 Check terminal junctions in Stateflow
Not Run

 Check for implicit type casting in Stateflow
Not Run

 Check usage of graphical functions in Stateflow
Not Run

 Check if state action type 'exit' is used in the model
Not Run

 Check for use of C-style comment symbols
Not Run

 Check usage of unconditional transitions in flow charts
Not Run

 Check for comments in unconditional transitions
Not Run

 Check definition of Stateflow events
Not Run

 Check Stateflow transition appearance
Not Run

 Check usage of events in Stateflow charts
Not Run

 Check usage of Simulink functions in Stateflow
Not Run

 Check for exclusive states in state machines
Not Run

 Check usage of floating-point expressions in Stateflow charts
Not Run

 Check use of Simulink in Stateflow charts
Not Run

 Check for arrowhead size of transition lines
Not Run

 Check for size of junctions
Not Run

 Check for overlapping of states and transition lines
Not Run

 *MATLAB Functions*  0  0  0  0  0  0  18

 Check input and output settings of MATLAB Functions
Not Run

 Check MATLAB code for global variables
Not Run

 Check usage of character vector inside MATLAB Function block
Not Run

 Check usage of enumerated values
Not Run

 Check description of execution statements
Not Run

 Check for spaces in function or variable names
Not Run

 Check for operator precedence
Not Run

 Check relational operator usage
Not Run

 Check spaces in expressions
Not Run

 Check for usage of non-empty statements
Not Run

 Check for utilization of the return value of functions
Not Run

 Check function headers
Not Run

 Check description of conditional expressions
Not Run

 Check the number of function calls in MATLAB Function blocks
Not Run

 Check nested conditions in MATLAB Functions
Not Run

 Check usage of recommended patterns for Switch/Case statements
Not Run

 Check array indices
Not Run

 Check number of lines of functions
Not Run

 Clone Detection and Model Refactoring 0 0 0 0 0 0 2

 *Refactor to simplify bus element blocks*
Not Run

 *Identify clones from a linked library file*
Not Run

 Simulink Code Inspector compatibility checks 0 0 0 0 0 0 70

 *Check shared Embedded Code Dictionary configuration*
Not Run

 *Check code generation settings*
Not Run

 *Check data import and export settings*
Not Run

 *Check diagnostic settings*
Not Run

 *Check hardware implementation settings*
Not Run

 *Check math and data types settings*
Not Run

 *Check solver settings*
Not Run

 *Check for unconnected objects in the model*
Not Run

 *Check system target file setting*
Not Run

 *Check for usage of fixed-point instrumentation*
Not Run

 *Check for unsupported blocks*
Not Run

 *Check storage class for workspace variables*
Not Run

 *Check GetSet storage class for workspace variables*
Not Run

 *Check for sample times in the model*
Not Run

 *Check usage of Sources blocks*
Not Run

 *Check usage of Signal Routing blocks*
Not Run

 *Check usage of Math Operations blocks*
Not Run

 *Check usage of Signal Attributes blocks*
Not Run

 *Check usage of Logical and Bit Operations blocks*
Not Run

 *Check usage of Lookup Tables blocks*
Not Run

 *Check usage of User-Defined Function blocks*
Not Run

 *Check usage of Ports and Subsystems blocks*
Not Run

 *Check usage of Discontinuities blocks*
Not Run

 *Check usage of Sinks blocks*
Not Run

 *Check usage of Discrete blocks*
Not Run

 *Check usage of root Outport blocks*
Not Run

 *Check for unsupported Signal Conversion blocks automatically inserted at signals entering block input ports*
Not Run

 *Check usage of buses*
Not Run

 *Check for usage of synthesized local data stores*
Not Run

 *Check usage of global data stores*
Not Run

 *Check global data stores' name shadow*
Not Run

 *Check conditional input branch execution setting*
Not Run

 *Check usage of Stateflow blocks*
Not Run

 *Check usage of Stateflow charts*
Not Run

 *Check usage of Stateflow data*
Not Run

 *Check usage of Stateflow events*
Not Run

 *Check usage of Stateflow states*
Not Run

 *Check usage of Stateflow junctions*
Not Run

 *Check usage of Stateflow ports*
Not Run

 *Check usage of Stateflow transitions*
Not Run

 *Check usage of Stateflow graphical functions*
Not Run

 *Check usage of Stateflow truth tables*
Not Run

 *Check Loop unrolling threshold setting*
Not Run

 *Check destinations of If and Switchcase blocks*
Not Run

 *Check for root Outport blocks that have non-auto storage class*
Not Run

 *Check for Terminator blocks that connect to Model block outports*
Not Run

 *Check for unsupported propagation of initial condition values*
Not Run

 *Check data type replacement names*
Not Run

 *Check usage of MATLAB Function Blocks*
Not Run

 *Check usage of Data in MATLAB Functions*
Not Run

 *Check usage of Code in MATLAB Functions*
Not Run

 *Check MATLAB Code Analyzer messages*
Not Run

 *Check for multiple sample times in model used as a model reference target*
Not Run

 *Check Treat each discrete rate as a separate task setting*
Not Run

 *Check model for commented out blocks*
Not Run

 *Check model for void_void subsystems that use the same function name*
Not Run

 *Check n-D Lookup Table blocks for incompatible breakpoint data type*
Not Run

 *Check model for reusable subsystems that use the same function interfaces*
Not Run

 *Check for usage of shared synthesized local data stores*
Not Run

 *Check for unsupported Code Mapping settings*
Not Run

 *Check model for compiled and graphical block sorted order*
Not Run

 *Check usage of String blocks*
Not Run

 *Check usage of shared utilities*
Not Run

 *Check model arguments for storage classes*
Not Run

 *Check usage of Stateflow MATLAB action language*
Not Run

 *Verify 'Generate an example main program' setting of multirate models*
Not Run

 *Check for unsupported usage of signals in Code Mappings editor*
Not Run

 *Check usage of Stateflow atomic subcharts*
Not Run

 *Check for unsupported usage of 64-bit data types*
Not Run

 *Check for unsupported usage of string data type*
Not Run

 Frequency Response Estimation 0 0 0 0 0 0 1

 *Identify time-varying source blocks interfering with frequency response estimation*
Not Run

 Simulink Design Verifier Compatibility Check 0 0 0 0 0 0 1

 *Check compatibility with Simulink Design Verifier*
Not Run

 Simulink Design Verifier Defect Checker 0 0 0 0 0 0 1

 *Check for common defects using Simulink Design Verifier Defect Checker*
Not Run

 Simulink Design Verifier Design Error Checks 0 0 0 0 0 0 13

 *Check for dead logic*
Not Run

 *Check for out of bound array access*
Not Run

 *Check for division by zero*
Not Run

 *Check for integer overflow*
Not Run

 *Check for non-finite and NaN floating-point values*
Not Run

 *Check for subnormal floating-point values*
Not Run

 *Check for specified minimum and maximum value violations*
Not Run

 *Check for data store access violations*
Not Run

 *Check for block input range violations*
Not Run

 *Check usage of remainder and reciprocal operations - hisl_0002*
Not Run

 *Check usage of square root operations - hisl_0003*
Not Run

 *Check usage of log and log10 operations - hisl_0004*
Not Run

 *Check usage of Reciprocal Square Root Blocks - hisl_0028*
Not Run

 Requirements Consistency Checking 0 0 1 0 3 0 0

 *Identify requirement links with missing documents (27-Apr-2026 23:27:54)*
Passed

 *Identify requirement links that specify invalid locations within documents (27-Apr-2026 23:27:54)*

Inconsistencies:

✔ *Identify selection-based links having description fields that do not match their requirements document text (27-Apr-2026 23:27:54)*

Passed

✔ *Identify requirement links with path type inconsistent with preferences (27-Apr-2026 23:27:54)*

Passed

📁 Units Inconsistencies 🚫0 ❌0 ⚠️0 🔄0 ✔️0 ⓘ0 📄5

📄⬇️ *Identify unit mismatches in the model*

Not Run

📄⬇️ *Identify automatic unit conversions in the model*

Not Run

📄⬇️ *Identify disallowed unit systems in the model*

Not Run

📄⬇️ *Identify undefined units in the model*

Not Run

📄⬇️ *Identify ambiguous units in the model*

Not Run